

Non-Fixed CT LightSpeed Systems Specs & Installation Requirements

The information in this document applies to:

- GRE Console
- GOC 1 and GOC2 Consoles
- HP60 Gantry Systems

OPERATING DOCUMENTATION



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Revision History

Revision	Date	Reason for change
0	09/27/2001	Initial release of document.
1	12/16/03	Added information for GRE console, GOC 1 and 2 consoles, H16 Gantry system, and the H3 Gantry system
2	07/08/04	Chapter 3: Updated Table 3-2 (PDU), Fig 3-2 (air vent locations) Chapter 4: Updated Section 1 (room temps) & 3 (added UPS) Chapter 5: Updated Section 1 & 2 mounting information Chapter 7: Corrected Section 2.5 voltage rating Chapter 8: Added UPS catalogs, updated Section 2 cable lists and interconnect diagrams, added NGDPU to Section 3.
3	9/08/05	- Throughout: Updated for HP60. Removed references to CRT monitors, except for those for Advantage Workstation. - Cover page: Deleted reference to this document applying to H16 and H3 gantry systems. - Chapter 2: Changed gantry motor assembly weight to 90 lbs. Deleted option listings of "International Dolly Set", "Remote Color Monitors", and "Long Cable Set". - Chapter 3: Revised dimensions for PDU. Updated Typical Scatter Survey diagrams. - Chapter 4: Revised cooling requirements for LCD monitors and deleted Advantage Windows option. Added section for Medical Electrical Equipment for EMC.
4	4/17/07	Chapter 4: Deleted references to static/fixed/specific clearances from scanner to PDU and/or UPS, basing component spacing instead on ambient magnetic fields.

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IMPORTANT PRECAUTIONS

LANGUAGE

警告 (ZH-CN)	• 本维修手册仅提供英文版本。 • 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。
VÝSTRAHA (CS)	• Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	• Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	• Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	• This Service Manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

HOIATUS (ET)	• Käesolev teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	• Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	• Ce manuel de service n'est disponible qu'en anglais. • Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris. • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	• Diese Serviceanleitung existiert nur in Englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	• Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS (HU)	- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	- Þessi þjónustuhandbók er eingöngu fáanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	- Il presente manuale di manutenzione è disponibile soltanto in inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.
警告 (JA)	- このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 . - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 . - 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 . - 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .
BRDINJUMS (LV)	- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu. - Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	• Šis eksploatavimo vadovas yra prieinamas tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė. • Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote. • Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.
ADVARSEL (NO)	• Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	• Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego. • Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT)	• Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.
ATENȚIE (RO)	• Acest manual de service este disponibil numai în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО! (RU)	• Данное руководство по обслуживанию предлагается только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.
UPOZORNENIE (SK)	• Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.
ATENCION (ES)	• Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	• Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DIKKAT (TR)	• Bu servis kilavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyariya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "Damage in Shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment, if not properly used, may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, GE Healthcare Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

IMPORTANT...RADIOACTIVE MATERIAL HANDLING

Only employees formally trained in radioactive materials handling and this equipment are authorized by the GE Healthcare Radiation Safety Officer to use radioactive materials to service this equipment.

GE Healthcare Services is required to notify the applicable U.S. state agency PRIOR to any source service event involving pin source handling. See NUC/PET Radioactive material guides for specific instruction or contact your EHS Specialist.

A radiation survey must be performed when a pin source has been removed and replaced. See Radiation Survey Form Instructions or contact your EHS Specialist.

Rev 2 (July 21, 2005)

OMISSIONS & ERRORS

Customers: please contact the GE Healthcare Sales or Service representatives.

GE personnel: please use the GE Healthcare iTrak/PQR Process to report all omissions, errors, and defects in this publication.

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Chapter 1

Introduction

Section 1.0

Expectations

This direction contains physical and electrical data necessary for planning and preparing a non-fixed unit for system installation. Non-fixed unit room layout, interior design, electrical system design, design for customer options and observation of all service clearances are ultimately the responsibility of the non-fixed unit manufacturer. Expectations can be summed as follow:

- The non-fixed unit manufacturer is responsible for all non-fixed unit designs, non-fixed equipment components, non-fixed unit construction and installation of the CT system in the non-fixed unit. GE is responsible only for GE equipment and its performance.
- Non-fixed unit designs must be certified by GE CT Engineering prior to any sales of said units.
- All non-fixed unit work must conform to all federal and local building and safety codes.
- Electrical system design is ultimately the responsibility of the non-fixed unit manufacturer, but should conform to the following:
 - Installation of electrical conduit, junction boxes, cable ducts or raceways, manufacturer supplied cables, manufacturer supplied outlets and line safety switches will be done using the most direct, point-to-point connection possible.
 - AWG stranded copper will be used in the installation of interconnection wiring. The non-fixed unit manufacturer will install the supplied GE mobile A1 panel. All wires shall be continuous without splices.
 - Ground wires must conform to local codes and still meet GE electrical grounding requirements.
- Unless specifically mentioned, GE will not:
 - Provide or install wires, conduits or ducts
 - Remove non-fixed unit manufacturer installed items
 - Relocate any junction boxes.
- The non-fixed unit manufacturer shall perform all equipment leakage tests between installed equipment and any conductive items in the unit. The test results will be communicated to GE in a written document with each unit.
- Any van/unit modifications, alterations or renovations will be communicated to GE through the existing process established between GE and the manufacturer.
- The non-fixed unit manufacturer will not perform any modifications to GE supplied products without first obtaining written approval from GE, unless the modifications are specifically included in the sales contract.
- The non-fixed manufacturer will observe all published service clearances and communicate any deviations to GE *prior* to implementation. If clearances cannot be met and new service tools are to be provided by the non-fixed unit manufacturer, then plans for these tools will be submitted to GE for review before work can continue.
- All CT non-fixed unit manufacturers and facility site plans, preliminary concepts and all working drawings must be reviewed by GE CT Engineering prior to construction or approval. This will be communicated to GE CT Engineering for final review before any production units are approved and design certified for construction.

- All non-fixed unit manufacturers will develop:
 - Procedures to train van company employees to mechanically install and connect the LightSpeed product, with options.
 - Instructions on how to mechanically install and connect the LightSpeed product, with options.
 - An installation completion check-list that outlines the process to store service equipment, including dollies, phantoms and manuals.
 - Procedures for the cleaning and touching-up of equipment.
 - Procedures for the storage of customer materials, including all non-fixed items, during transportation.

GE CT Engineering reserves the right to reject any installation training plan, mechanical installation plan or other plans and drawings that do not observe all service clearances and requirements.

Contact your non-fixed unit GE CT Engineering representative for concerns regarding your specific layout.

Section 2.0

Responsibilities

It is the responsibility of the unit manufacturer to communicate to the buyer all site requirements that are necessary to plan and prepare a site for receiving a van/unit.

- Determine and verify parking area dimensions are large enough to maintain the van.
- Determine and verify parking area is safe for use of the van.
- Communicate all power, network and telephone connector types to be installed at the buyer facility.
- Communicate to the buyer a list of any external cables, lights, safety lines or any other items they are expected to provide.
- Communicate to the buyer how to obtain replacement parts, such as lamps, controls or entertainment systems, as well as where to obtain service for all non-GE supplied equipment (such as HVAC, generator, lift, etc.).
- Provide the buyer with an Operator's Manual, Service Manual and training, if required to safely operate the manufacturer's non-fixed system.
- Discuss with the buyer any hazards, service clearance issues and other changes made to the manufacturer's non-fixed environment.

Chapter 2

System Catalog

Section 1.0

Option Catalog Numbers

The following is a list of system options requiring construction/structural planning work for Non-fixed LightSpeed systems. *Contact the local GE Healthcare Sales representative for a complete list of all system options or visit us at <http://www.gehealthcare.com>. Refer to the instruction manuals supplied with specific options for respective details.*

CATALOG NUMBER	OPTION DESCRIPTION
SmartScore Option B7850KC	EKG Monitor and Recording Device
ConnectPro Option B7500PL	ConnectPro Option provides a direct interface to HIS/RIS
Bar Code Reader B7540RB	LightSpeed 5.0 Bar Code Reader
Modem B7700MG	Global Modem Kit

Table 2-1 System Options Catalog (Part) Numbers

Section 2.0

Base Scanner System

The base scanner system is configured as shown. All scan and analysis functions are controlled from the operator's console (not shown).



Figure 2-1 Base Scanner System

Chapter 3

Site Planning

Section 1.0

Common Dimensions and Clearances

1.1 System Service

• Front cover removal (side clearance)	36.0" (914mm)
• Gantry service side to obstruction	36.0" (914mm)
• Area around the PDU	36.0" (914mm)
• Area around the UPS	36.0" (914mm)
• Area around the A1 breaker box	42.0" (1067mm)
• Area between the table down foot end to obstruction	36.0" (914mm)
• Minimum Gantry rear cover removal area	50.0" (1270mm)

1.2 System Operation

• Finished ceiling to floor	108.0" (2732mm)
• Table max extension head end with extender from Center Line	80.0" (2032mm)
• Table extension head end with extender to obstruction	6.0" (152mm)
• Table in lowest position w/cradle at home position to Center Line	126.5" (3213mm)
• Back of Console to wall - all sides	6.0" (152mm)
• Back of PDU to wall - all sides	6.0" (152mm)

Additional dimensions are available in [Figure 3-1](#) through [Figure 3-5](#) of this document.

Section 2.0

Injector and Injector Control

It is recommended that a suitable work area be provided for the placement of injector control that is within reach of the operator console.

Wall mounted, ceiling mounted and pedestal units need cables routed from the gantry area to the console area. The supplied cable is 50 feet long. Injectors require AC power. Mounts are available in different configurations and lengths. Refer to Injector documentation for detailed installation instructions. Injector placement information is included in this document.

Only mobile environment certified wall mounted or ceiling mounted injectors shall be installed in mobile units.

Section 3.0

Storage Cabinet

A storage cabinet will be provided by the unit manufacturer to store all supplied service equipment (see [Table 3-1](#) for equipment list). This storage cabinet should be located in an area that allows for easy service access.

All items listed below must be shipped stored in this cabinet, when the van is delivered. Shelves must be able to support the weights listed in the table below.

ITEM	SIZE		WEIGHT (TOTAL)
	CM	INCHES	
QA Phantom (water filled)*	20 x 15	7.9 x 5.9	12 lb
20CM Phantom	20 x 7	7.9 x 2.75	13 lb
48CM Phantom	48 x 7	18.9 x 2.75	25 lb
Phantom Holder	25 x 25	9.8 x 9.8	8 lb
Gantry Display	62 x 13	24.4 x 5.1	2 lb
FE Box (Documents)	30 x 38 x 30	11.8 x 15 x 11.8	15 lb (ea.)
Rear Cover Dollies**	158 x 82	62.2 x 32.3	25 lb (ea.)
Front Cover Dollies	85 x 20 and 85 x 15	33.5 x 7.9 and 33.5 x 5.9	35 lb
Install Support Kit (box)	30 x 30 x 38	11.8 x 11.8 x 15	20 lb
Tube Hoist Kit	77 x 8 and 38 x 15	30.3 x 3.1 and 15 x 5.9	20 lb
Gantry Control Pad	27 x 20	10.6 x 7.9	4 lb
Gantry Balance Weight Kit	-	-	40 lb
* Phantom will freeze, if temperature is below 32° F (0° C)			
** Dollies must be securely stored, hanging vertically.			

Table 3-1 Equipment to be stored in storage cabinet

Section 4.0

Advantage Windows Workstation (AWW)

- Refer to Pre-Installation Manual 2111833 and Installation/Service Manual 2111831.
- CRT monitors must be mounted on GE approved isolation panels.
Flat Panel monitors shall be mounted in the same fashion as the OC monitors.
- A mouse holder shall be supplied by the non-fixed unit manufacturer.

It is ultimately the responsibility of the non-fixed unit manufacturer to design appropriate structural mounting for ceiling and wall mounted injectors.

Section 5.0

UPS

The Mobile UPS system is shipped with each mobile van. The mobile UPS consists of one main unit, which weighs 500 pounds, and an auxiliary contactor box with cables. Detailed installation instructions are enclosed with the mobile UPS.

It is ultimately the responsibility of the non-fixed unit manufacturer to design appropriate structural mounting for ceiling and wall mounted injectors.

Section 6.0

Replacement Component Dimensions

The following are major system components that will be expected to be field replaceable on all non-fixed systems.

6.1 X-Ray Tube

29" H x 28" W x 48" L 275 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

6.2 High Voltage Tank

Single: 23" H x 16" W x 14" L 35 lbs

Dual: 23" H x 27" W x 17" L 70 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front right side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

6.3 Detector

17" H x 36" W x 55" L 90 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front right side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

6.4 DAS

27" H x 16" W x 27" L 40 lbs

Shipped in a box. Must have unobstructed path to move this box to the front of the gantry and also have working room around the gantry. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

6.5 Slip Ring

60" H x 60" W x 10" D 128 lbs

Shipped in wooden box. Must have unobstructed path to move this box to the front right side of the gantry and also have working room on the gantry right side. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

6.6 Gantry Motor Assembly

17" H x 12" W x 30" D 90 lbs

Shipped in a box. Must have unobstructed path to move this box to the front of the gantry and also have working room around the gantry. Must have room to remove the front and rear covers, using the current cover dollies. Must have ceiling height of 84" to operate hoist and boom, installed on the gantry right front side.

6.7 HP Computer

24" H x 26" W x 17" D 50 lbs

Shipped in a box. Must have unobstructed path to the console. Must have room to remove the front of the console.

6.8 IG

12" H x 30" W x 30" D 20 lbs

Shipped in a box. Must have unobstructed path to the console. Must have room to remove the front of the console.

6.9 DARC

12" H x 30" W x 30" D 30 lbs

Shipped in a box. Must have unobstructed path to the console. Must have room to remove the front of the console.

6.10 SDDA

12" H x 30" W x 30" D 25 lbs

Shipped in a box. Must have unobstructed path to the console. Must have room to remove the front of the console.

6.11 Recon Unit

12" H x 12" W x 31" D 35 lbs

Shipped in a box. Must have unobstructed path to the console. Must have room to remove the front cover of the console.

Section 7.0

Component Dimensions

DESCRIPTION	WIDTH INCH (MM)	DEPTH INCH (MM)	HEIGHT INCH (MM)
System			
Gantry	86.6 (2200)	39.4 (1000)	74.6 (1895)
Table	24.3 (617)	94 (2387)	
Power Distribution Unit (PDU)	28 (711)	22 (559)	42 (1067)
Operator's console/computer	48 (1219)	39 (991)	33.5 (851)
Color printer*	23 (584)	18 (457)	7 (178)
Workstation	16.5 (419)	16.1 (409)	3.1 (79)
UPS	30 (762)	16 (406)	28 (711)
A1	20 (508)	8 (203)	31 (787)

* Must have shock platform for this option.

Table 3-2 Dimensions of Components

7.1 Table and Gantry

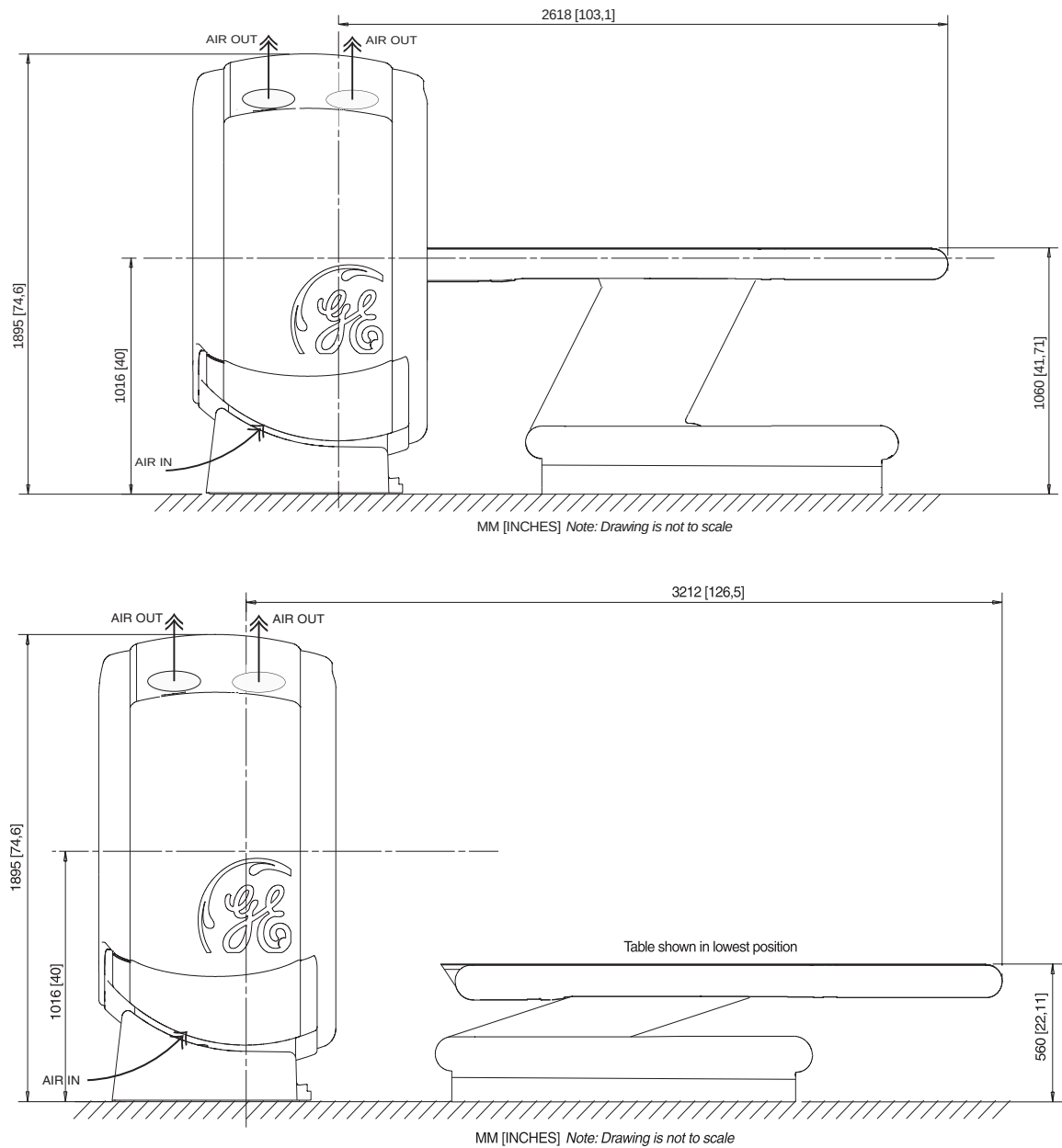


Figure 3-1 Table and Gantry (Side View)

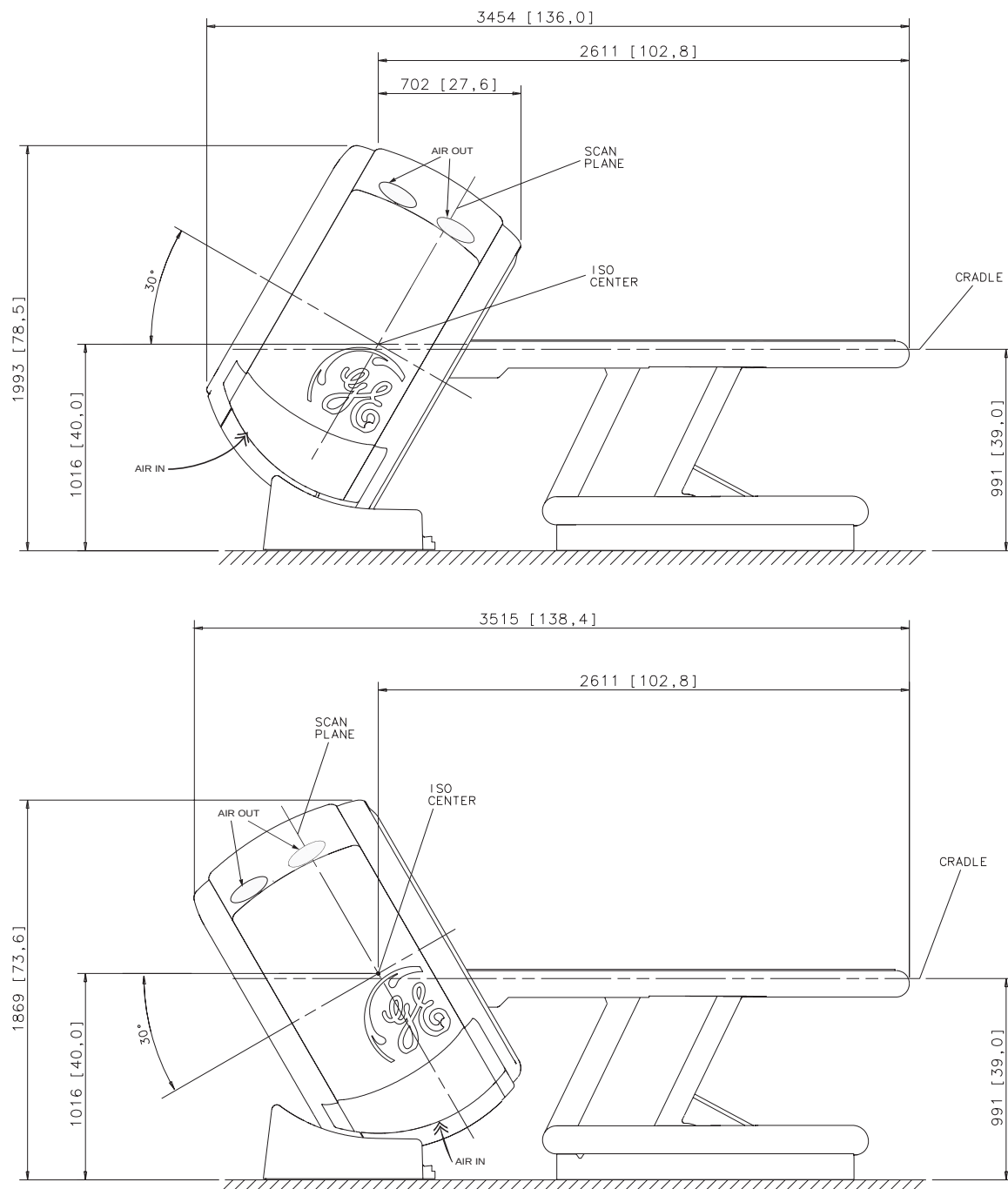


Figure 3-2 Gantry shown tilted +30° (top) and -30° (bottom)

7.2 Power Distribution Unit

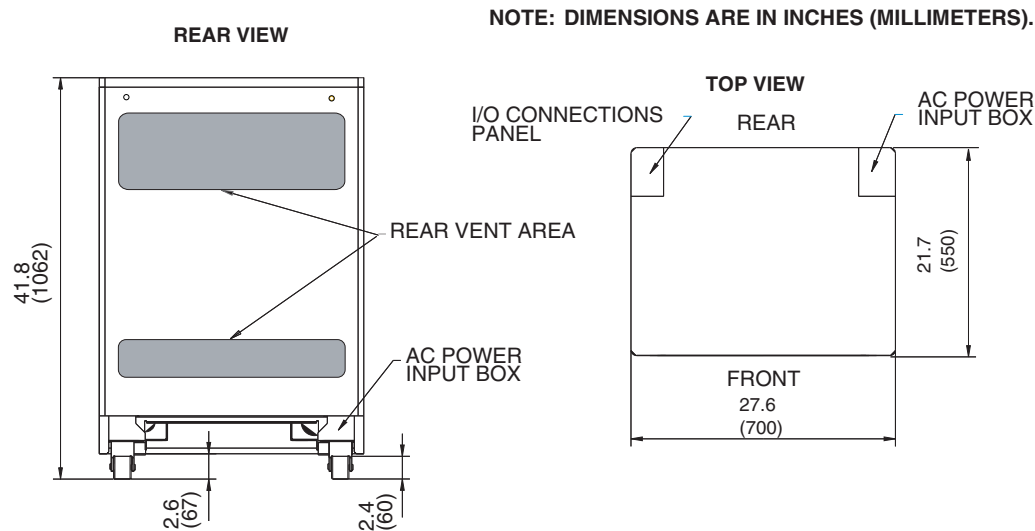


Figure 3-3 Power Distribution Unit (NGPDU)

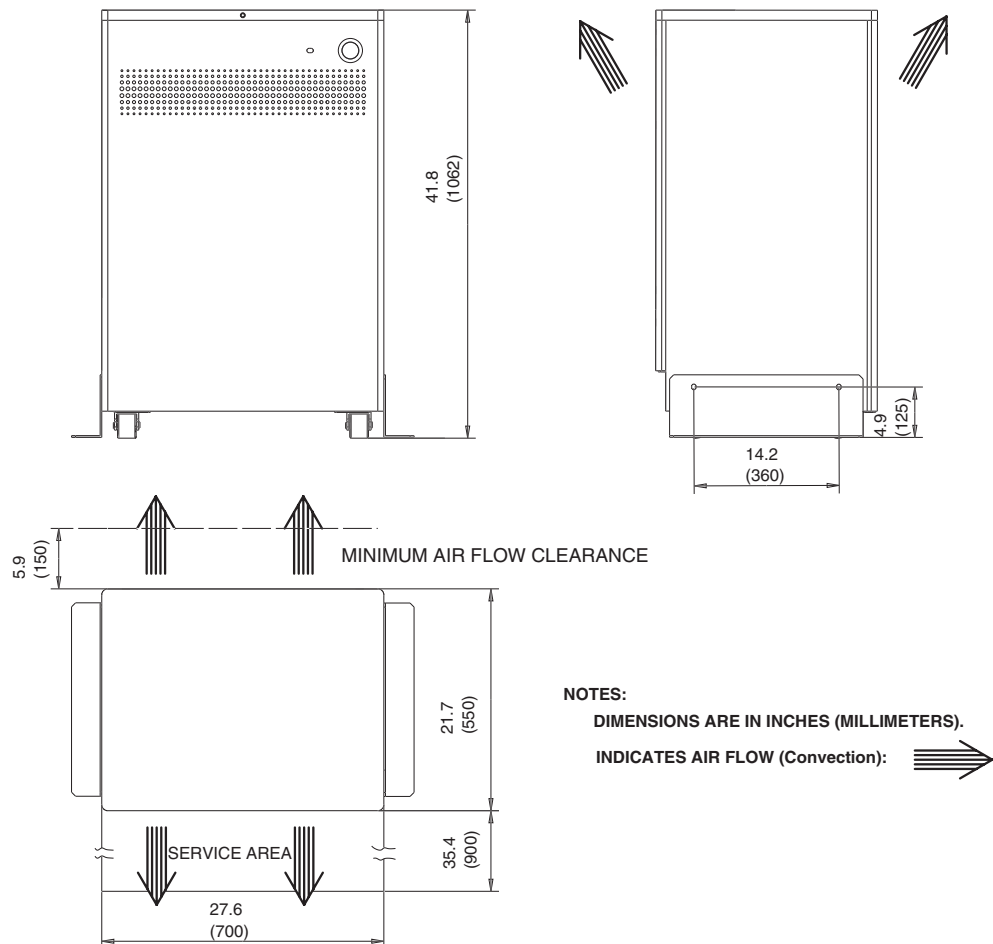


Figure 3-4 Power Distribution Unit (NGPDU)

7.3 Operator's Console

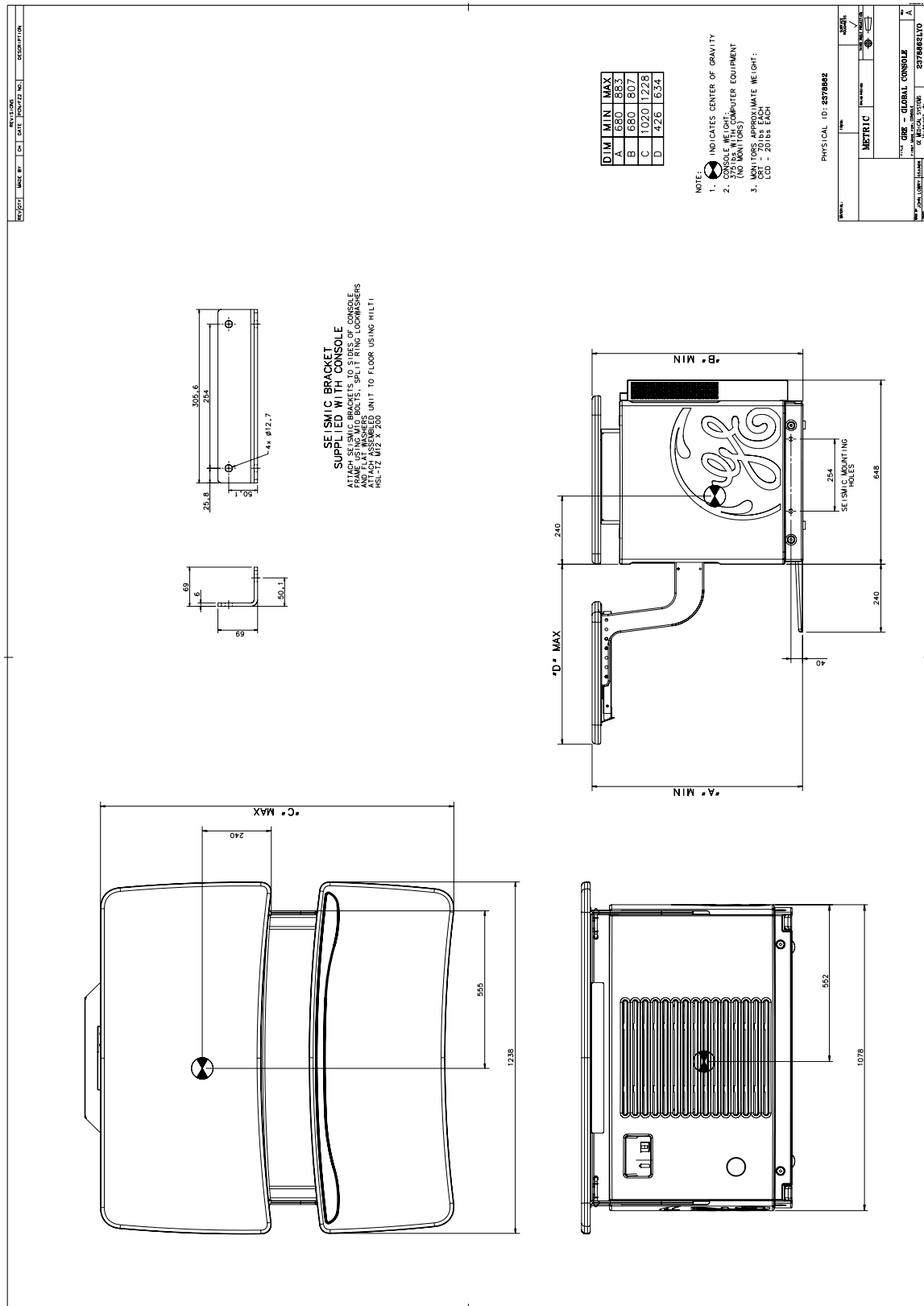


Figure 3-5 Operator's Console

Section 8.0

Structural Requirements

It is ultimately the **responsibility of the non-fixed unit manufacturer** to design the van/unit solution (i.e., levelness, vibration, equipment location, and proximity effects of surrounding equipment such as other mobile, transportable, and relocatable vans that could contain CT, MR, or other medical devices).

8.1 Suggested Ceiling Heights

A ceiling height of 108" (2743mm) is recommended. A ceiling height of 92" (2337mm) is the absolute minimum acceptable height for gantry installation.

8.2 Table and Gantry Mounting Requirements

 **WARNING** **POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

 **NOTICE** It is the non-fixed unit manufacturer's responsibility to provide an approved support structure and mounting method. Gantry mounting requires $\frac{3}{4}$ " structural bolts, and $\frac{1}{2}$ " structural bolts are required for table mounting. Furthermore, $\frac{1}{2}$ " steel shim plates placed under the gantry (refer to [Figure 5-3, on Page 51](#)) are required for all vans. This steel shim is not required for non-fixed relocatable buildings. All gantry and table mounting must be centered to the drilled holes, to allow for X-Ray to table alignment. A drilling centering tool will be provided by GE to all non-fixed unit manufacturers. GE is not responsible for any failure of the support structure or method of anchoring.

Table and gantry mounting dimensions are shown in [Figure 3-1](#). Refer to Chapter 5 for additional details of floor loadings, component weights, and Gantry and Table installation and anchoring.

Anchor gantry and table to floor by a means that will maintain their relative alignment and meet applicable building and other local codes.

Floor structure must be capable of withstanding the occupied weight of table and gantry, and the individual contact area loading of these components. **The table-gantry requires a floor support in the 100 lbs./sq. ft. (490 Kg/sq m) class or above. Any installation on a floor with a rating less than 100 lbs./sq. ft. (490 Kg/m²) should be braced to bring it up to this requirement.** Localized bracing to support the concentrated loads at the floor contact sections should also be provided.

Support areas of the patient table and gantry must rest on the steel flooring, not resilient tile or carpeting that will slowly yield over a period of time and disturb alignment of table to gantry. Use the supplied template to locate the areas where this floor material should be removed prior to installing the table or gantry. There should be ten (10) locations on the gantry and five (5) on the table.

Factors that could cause misalignment between gantry and table due to floor sag should be considered. The cradle can potentially carry a 450 lb (205 kilogram) patient. Center of gravity changes as cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to Chapter 5 for gantry and table mounting details.

No part of floor surface within table and gantry, nor the two interface areas between table and gantry, should be higher than the support area for table and gantry.

The non-fixed unit manufacturer is responsible for engaging a qualified person to verify that the non-fixed unit and method of anchoring are adequate to support loads and maintain table-to-gantry alignment. Location of supporting beams and columns may dictate position of table-to-gantry assembly. Use of flush floor duct or conduit in the floor may significantly affect floor strength.

The method and placement of anchoring through bolts must not reduce structural strength of floor.

8.3 Floor Strength

Van floors must have a minimum strength of $f'_c = 2000$ psi (1.4×10^7 Pa) at 28 days (concrete spec) for mounting floor anchors. It is the responsibility of the non-fixed unit manufacturer to have appropriate tests performed to meet these specifications.

8.4 Floor Levelness

Floor levelness in the Scan Room must not be greater than 0.3125 (8mm) between depression and high spots over any 120 in. (3048 mm) distance within the area of the gantry and the area around the table. Means to accurately level the non-fixed unit must be provided by the non-fixed unit manufacturer.

8.5 Floor Vibration

The CT equipment is sensitive to vibration in the frequency range of 0.5 to 20 Hz, depending on the amplitude of the vibration. It is the responsibility of the van manufacturer to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits. However, it is ultimately the responsibility of the facility that uses the non-fixed unit to design the *site* solution.

8.5.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s² rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period). Operating at this specification, diagnostic image quality inside the non-fixed unit will not be diminished.

8.5.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s² peak-to-peak. Operating at this specification, diagnostic image quality inside the non-fixed unit will not be diminished.

8.5.3 Operation in High Winds

The behavioral characteristics of the non-fixed unit must be such that allows the owner to operate the systems in winds up to 45 MPH without any change to diagnostic image quality inside the non-fixed unit.

8.6 Scan Window

For patient safety, the operator at the Operator Workspace must be able to view the patient during a scan. The non-fixed unit suggested viewing window dimensions are 36 in. wide by 42 in. high (914 mm x 1067 mm).

Section 9.0 Network Connections

Broad-band is considered the standard network connection for LightSpeed 5.X. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the facility's network through the Console or router.

- A patch cable (not to exceed 10 feet) provided by the non-fixed unit manufacturer, is used to connect the console to a wall box or optional router. (See Notes on [Figure 8-4](#))
- The facility or non-fixed unit manufacturer will need to provide applicable weather rated network connections from the facility to the van.
- Some non-fixed units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- Non-fixed unit manufacturer will provide an under-counter shelf to mount a network router box.
- The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice only line.

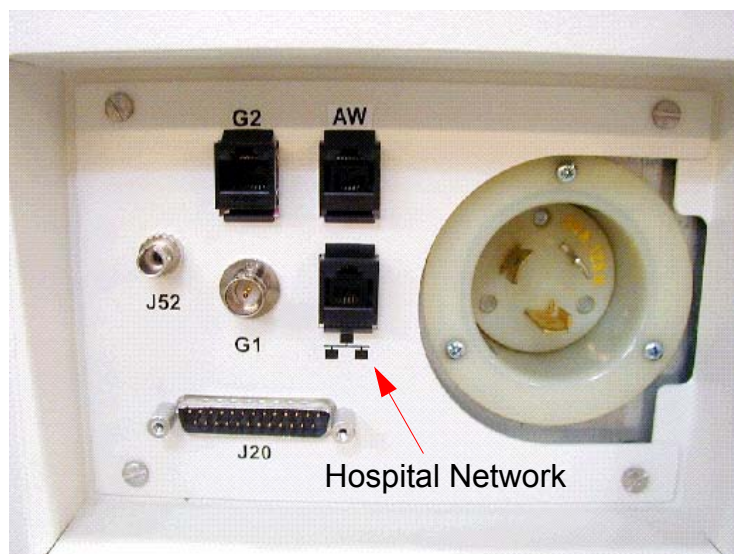


Figure 3-6 Console Rear Bulkhead

9.1 US Process Overview

The United States network connectivity requirement for this product is broad-band. The US process relies on the Install Specialist to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

9.2 Customer Broad-Band Responsibilities

Provide GEMS Installation Specialist with an accurate site address, telephone number, contact name, and email address for the:

- Customer Champion
 - Co-ordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broad-band infrastructure meets GEMS requirements for connection as determined by a mutual assessment with the GEMS Connectivity team.
- IT Contact
 - Complete an equipment assessment with GEMS Connectivity team to determine site readiness for broad-band
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - To utilize an Internet Service Provider that supports static routing



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ FE Name: _____
 City, State: _____ FE Phone: _____
 Date: _____ FE Email: _____

- | | Yes | No |
|--|---|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |
| ☐ a) Cisco Pix Firewalls | ☐ g) Symantec (Raptor) firewalls | |
| ☐ b) Cisco Routers | ☐ h) Firebox | |
| ☐ c) Cisco 3000 Series (Altiga acquisition) | ☐ i) Linux S/WAN | |
| ☐ d) Checkpoint Firewalls Software Version 4.1 and higher | ☐ j) Sidewinder | |
| ☐ e) Nortel Contivity Software Version 3.2 or higher | ☐ k) Netscreen | |
| ☐ f) Redcreek | ☐ l) None | |
| | ☐ m) Other _____ | |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

5. Does your VPN device support "triple DES" Encryption? ☐ Yes ☐ No
6. Has approval been given to install this VPN connection? ☐ Yes ☐ No
- Site Approver's Name _____
7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
 Installer Telephone Number: _____
 Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
 Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:

a) Judy Heyer judy.hey@med.ge.com
 b) Judy Heyer Fax# 414-918-4707

[Use the send button on page 2](#)



GE Medical Systems

System ID List & Compatibility Matrix

MR	Software Version	CT	Software Version
Include all Signa(LX, Ovation, OpenSpeed and CV/I) MR's System Ids and IP Addresses in assessment.		HiSpeed CT/e	Minimum 6.0
		HiSpeed CT/e dual	Minimum 2.06
		HiSpeedNX/I	Minimum 5.5
		LightSpeed QX/I	Minimum 1.3
Prior to completing an InSite checkout systems must be upgraded to Version 9.0 or 9.1 unless shown in the compatible list below.		LightSpeed Plus	Minimum 2.1
		LightSpeed Ultra	Minimum 3
		HiSpeed X/I	
		CT/I	Minimum 6.2
Signa TwinSpeed	Release 9.0	Lxi, Dxi, Fxi, Zxi	Minimum 6.01
Signa	Release 9.1		
Signa	Release 10.x		
Signa CV/I	CNV4		
Signa 3T			
Signa Profile	Minimum 7.66		
Signa Contour	Minimum 7.66		
Magnet Monitor	Minimum 2.3 software		

Network Products	Software Version	Nuclear/Pet	Software Version
AW 4.0	Version 4.0 or Above	Advance	Minimum 5.1
AW 4.0P		eNTEGRA	Minimum 2.03

X-Ray	Software Version
<i>Cardiac</i>	
INNOVA	All
<i>Mammography</i>	
Seno2000D	All
<i>Digital Radiology</i>	
Revolution XQ/i	Version 10.12.5 or above
Revolution XR/d	Version 18.0

Multi Vendor

MR	PSI	CT	PSI
Philips ACS 1.5T	MZP 400	Picker MX-TWIN	CZM 400
Philips NT 2000	MZP 401	Picker MX 8000	CZM 401
Philips NT 3000	MZP 402	Picker PQ	CZM 500
Philips NT ACS	MZP 403	Picker PQ 2000	CZM 501
Picker Outlook 0.23T	MZM 301	Picker PQ 5000	CZM 502
Picker Edge 1.5T	MZM 800	Picker PQ 6000	CZM 503
Picker Vista 1.0T	MZM 801	Picker PQS	CZM 504
Picker Eclipse 1.5T	MZM 900	Siemens AR STAR	CZS 100
Picker Polaris 1.0T	MZM 901	Siemens AR.C	CZS 101
Siemens Harmony 1.0T	MZS 100	Siemens AR.HP	CZS 102
Siemens Impact 1.0T	MZS 200	Siemens AR.SP	CZS 103
Siemens Impact Expert 1.0T	MZS 201	Siemens AR.T	CZS 104
Siemens SP 1.0T	MZS 400	Siemens PLUS-4 EXP	CZS 400
Siemens SP 1.5T	MZS 401	Siemens PLUS-4 POW	CZS 401
Siemens SP4000 1.5T	MZS 403	Siemens PLUS-4 VOL	CZS 402
Siemens Symphony 1.5T	MZS 500		
Siemens Vision 1.5T	MZS 600		

Field Engineer needs to provide compatible system information:

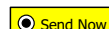
[illegible]

Once you have completed this form, please send it to:

Judy Heyer
Judy Heyer

email
Fax #

judy.heyer@med.ge.com
414-918-4707



Section 10.0

Radiation Protection

10.1 Non-Fixed Unit Manufacturer Shielding Requirements

Scanner-room shielding requirements should be reviewed by a **qualified radiological health physicist** taking into consideration equipment placement, weekly projected work-loads, and materials used for construction of walls, floors, ceiling, doors, and windows.

- It is the non-fixed unit manufacturer's responsibility to provide approved radiation shielding in all areas of the environment.
- Non-fixed units must meet all approved and acceptable radiation leakage requirements.

Figure 3-7 depicts measurable radiation levels within the scanning room while scanning a 32 cm CTDI phantom (body) and a 20 cm water phantom (head) with the technique shown.

Note:
Actual
measurements
can vary.

All measurements have an accuracy of $\pm 20\%$ because of measurement equipment, technique, and system-to-system variation.

Use the correction factors shown in Figure 3-3 to adjust exposure levels to the usual scan technique at the site.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/100
80 kV	0.21
120 kV	0.71
140 kV	1.0
4 x 3.75mm images	0.82
16 x 0.625 LD	0.59
8 x 1.25 LD	
4 X 2.5 LD	
Fluro 5mm	
4 x 1.25 LD	0.40
5mm (1i)	
Fluro 2.5 mm	
1 x 1.25mm images	0.20
2 x 0.625 LD	0.10
1 x 1.25	

Table 3-3 Shielding Requirements Scaling

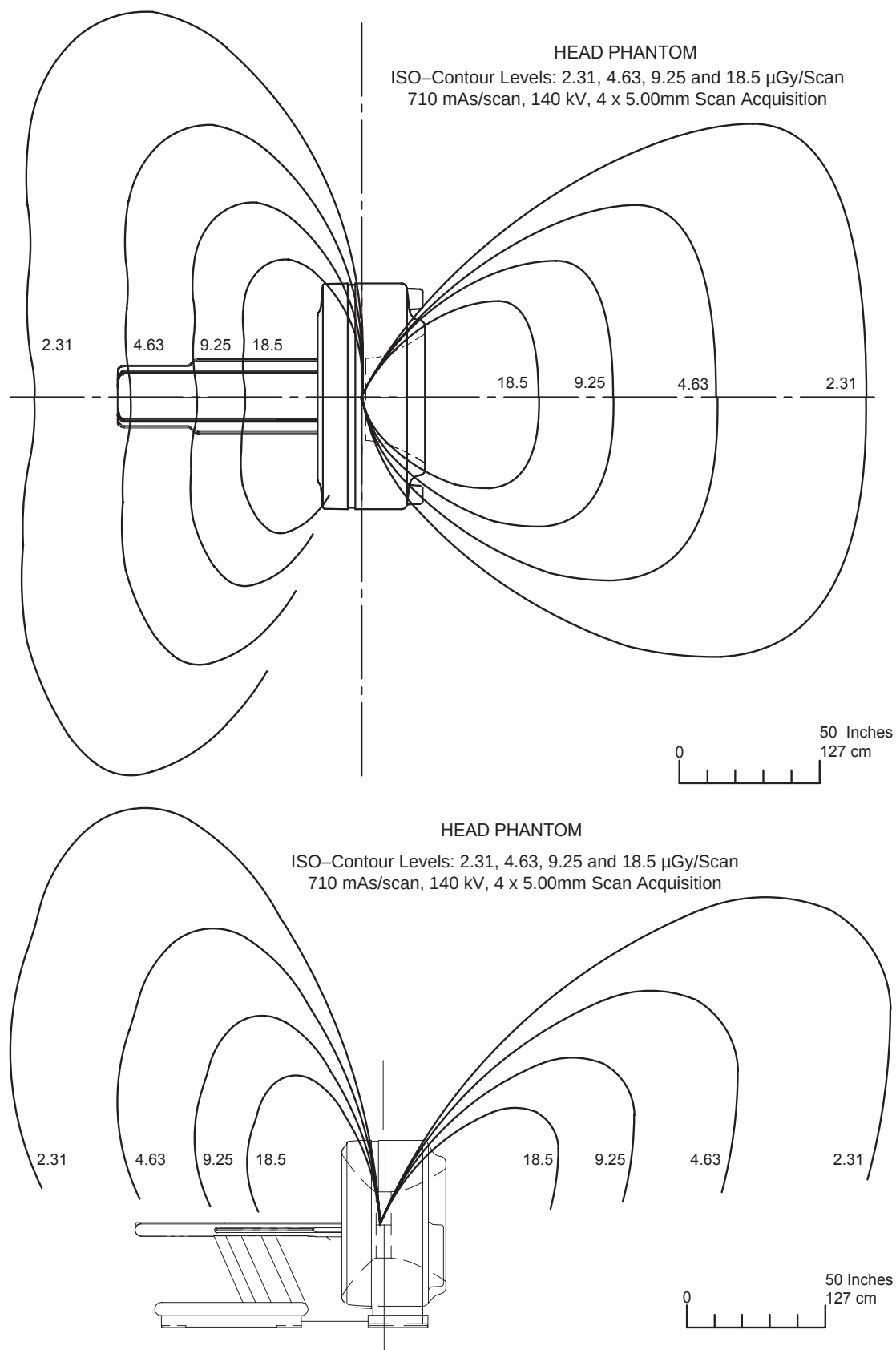


Figure 3-7 Typical Scatter Survey (Head Filter)

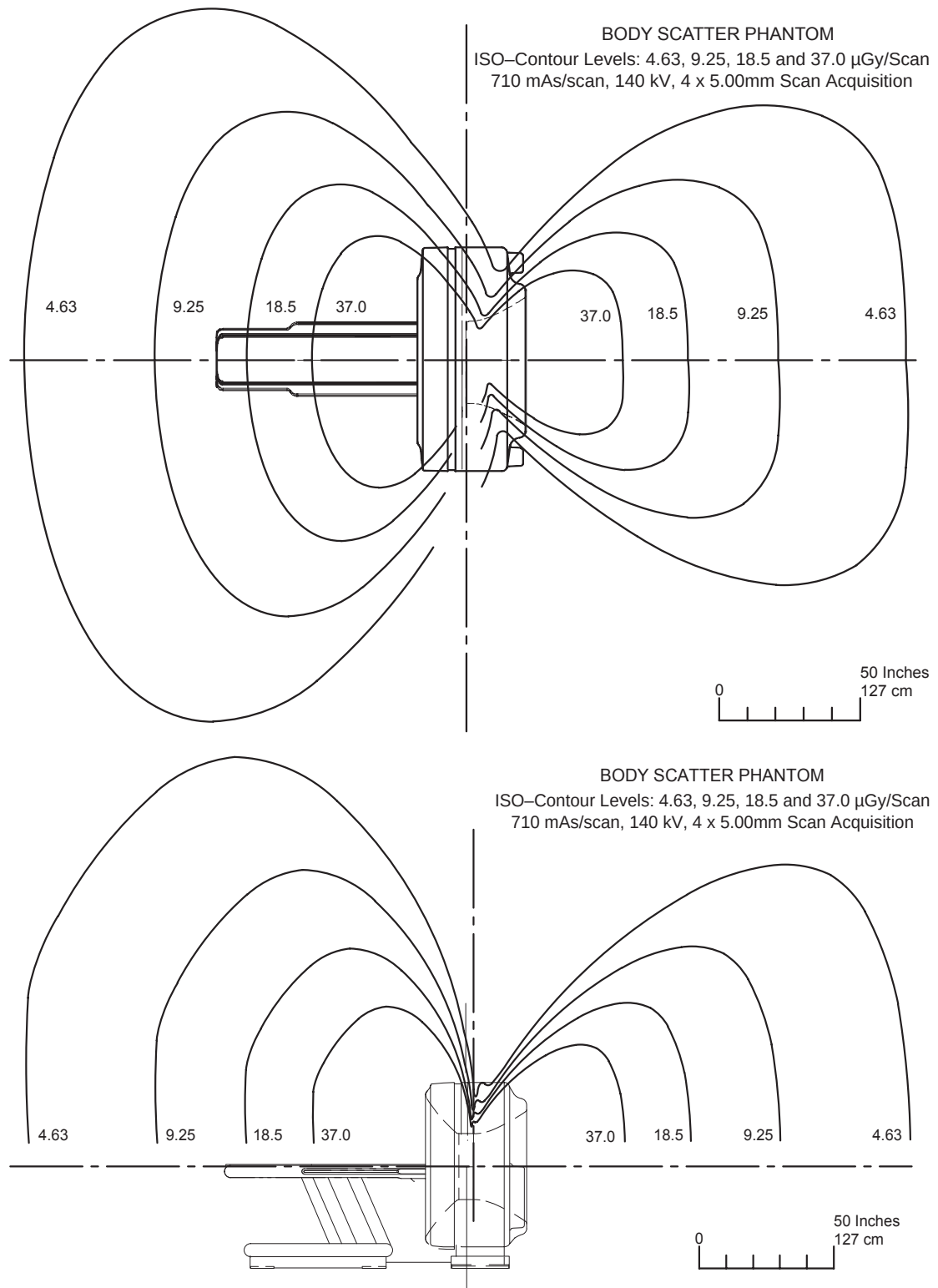


Figure 3-8 Typical Scatter Survey (Body Filter)

10.2 Facility to Use Mobile Van

It is ultimately the responsibility of the facility to provide the mobile site solution.

Chapter 4

Environmental Conditions

- Ratings and duty cycles of CT subsystems apply if non-fixed unit manufacturer meets the environment standards of this section.
- It is the van manufacturer's responsibility to maintain environmental conditions listed below at all times – including, for example, overnight, weekends, holidays and storage.
- Shut down the CT system if air conditioning is not working. When system is shut down for major repair, air conditioning may be shut down also.

Section 1.0

Temperature and Humidity Specifications



NOTICE
Potential
Equipment
Failure

Do not operate (i.e., “Power ON”) Gantry or Console subsystems with ambient room temperatures exceeding 75° F. Scan room or control room temperatures in excess of 75° F can result in the failure of gantry or console components.

- Ambient Temperature: (Fahrenheit and Celsius)
 - Scan Room: Maintain a temperature of 72° F $\pm$ 3° F (22° C), for patient comfort. When scan room is unoccupied, table and gantry temperature limitations are 70° - 75° F (21° - 24° C).
 - Control Room (including Console/Computer): Maintain 72° F $\pm$ 3° F (22° C).
 - Equipment Room: If a separate equipment room is used to house the PDU, the allowable temperature range is 60° - 75° F (15° - 24° C).
- Store media (cartridges) in long-term storage in same temperature range as host computer.
- Store media in the host computer environment for one-half hour before use.
- Maintain relative humidity of 30%-60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 5° F/hr. (3° C/hr.).
- The maximum relative humidity rate of change is 5% RH/hr.

Section 2.0

Temperature and Humidity Monitoring

Locate computing subsystems in an area that has the environment specified in [Section 1.0](#), above. First, assess the environment's heat and humidity. If necessary, temporarily install a temperature and humidity recorder close to where the gantry will be installed. Note readings before installation and again after installation to verify true temperature and humidity conditions for the environment. Consider the HVAC and redundancy needs. An air conditioner with two compressor units may be more appropriate than one. A backup (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 3.0

Cooling Requirement

Use [Table 4-1](#) to assist in cooling requirements planning. Over half the cooling used by the scanner is required for gantry operation. For patient comfort, locate a wall air-conditioning vent at floor level beside and behind gantry to meet both gantry cooling needs and provide patient comfort.

SYSTEM COMPONENT	BTU/HR	WATT
1. Gantry recommended (See NOTE 1)	25,100	7,350
2. Table	700	200
3. Power distribution unit	3400	1000
Recommended Scan Room Subtotal (see notes):	29,200	8,550
4. Operator's console/computer with one IG	7361	2165
Additional IG, each	1360	400
Monitor, each (LCD)	425	125
SCSI Tower	425	125
Injector	425	125
DASM	425	125
Recommended Control Room Subtotal (includes OC w/1 IG, 2 LCD monitors & SCSI Tower):	12,342	3630
System Total (Recommended) (See NOTE 1)	41,542	12,180
Option: UPS	2500	732
Option: Advantage Windows	256	75
ROOM TOTAL (SEE NOTE 2)		
NOTE 1: With 75 scan rotations per patient: Recommended BTU/hr. provides for up to six patients per hour. It is also needed during calibration of the system.		
NOTE 2: Cooling requirements do not include cooling for room lighting, personnel or non-CT equipment.		

Table 4-1 Cooling Requirements (Worksheet)

Section 4.0

Altitude

System operating altitude is from mean sea level to 10,000 ft. (3050 meters).

Section 5.0

Medical Electrical Equipment for EMC

5.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2 EMC standard for medical electrical equipment.

The **EQUIPMENT and/or SYSTEM** System is suitable to be used in the electromagnetic environment, as per the limits and recommendations described in the tables hereafter:

- Emission Compliance level and limits (See [Table 4-2](#).)
- Immunity Compliance level and recommendations to maintain equipment clinical utility (See [Table 4-3](#), [Table 4-4](#), and [Table 4-5](#).)

Note: This system complies with above mentioned EMC standard when used with supplied cables up to maximum lengths referenced in the MIS MAPS or system cable interconnect diagrams.

5.2 Electromagnetic Emission

Table 4-2 Emission Declaration

EMC Emissions Guidance & Declaration for EQUIPMENT and/or SYSTEM		
The EQUIPMENT and/or SYSTEM is intended for use in the electromagnetic environment specified below. The customer or the user of the EQUIPMENT and/or SYSTEM should assure that it is used in such an environment.		
Emissions Test	Compliance	Electromagnetic Environment Guidance
RF emissions CISPR 11	Group 1	The EQUIPMENT and/or SYSTEM uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	(Phase 1 - only) Class A + 12 (Phase 2 - only) Class A	(Phase 1 only) <i>Warning:</i> This equipment is allowed to be installed only in X-ray protected rooms, which provide an attenuation of at least 12 dB for radio disturbances from 30 MHz to 1 GHz. When installed in such a shielded location:
Harmonic emissions IEC 61000-3-2	Not applicable	The EQUIPMENT and/or SYSTEM is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Voltage fluctuations/ flicker emissions IEC 61000-3-2	Not applicable	

5.3 Electromagnetic Immunity

Table 4-3 Immunity Declaration

EMC Immunity Guidance & Declaration for EQUIPMENT and/or SYSTEM			
The EQUIPMENT and/or SYSTEM is intended for use in the electromagnetic environment specified below. The customer or the user of the EQUIPMENT and/or SYSTEM should assure that it is used in such an environment.			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input/output lines	± 2 kV for power supply lines ± 1 kV for input/output lines	Mains power quality should be that of typical commercial or hospital environment.
Surge IEC 61000-4-5	± 1 kV line-line ± 2 kV line-earth	± 1 kV line-line ± 2 kV line-earth	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	< 5 % U_T (> 95% dip in U_T) for 5 sec	< 5 % U_T (> 95% dip in U_T) for 5 sec	Mains power quality should be that of a typical commercial or hospital environment. If the user of the EQUIPMENT and/or SYSTEM requires continued operation during power mains interruptions, it is recommended that the EQUIPMENT and/or SYSTEM be powered from an uninterruptible power supply or a battery.
Powerfrequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
Note: U_T is the a.c. mains voltage prior to application of the test level.			

Table 4-4 Immunity Declaration, cont.

EMC Immunity Guidance & Declaration for EQUIPMENT and/or SYSTEM			
The EQUIPMENT and/or SYSTEM is intended for use in the electromagnetic environment specified in this table. The customer or the user of the EQUIPMENT and/or SYSTEM should assure that it is used in such an environment.			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Conducted RF IEC 61000-4-6	3 V _{RMS} 150kHz to 80 MHz	3 V 150kHz to 80 MHz	Portable and mobile RF communications equipment should be used no closer to any part of the EQUIPMENT and/or SYSTEM, including cables, than the recommended separation distance calculated from the equation appropriate for the frequency of the transmitter. Recommended Separation Distance $d = \left[\frac{3.5}{3} \right] \sqrt{P}$ (See Table 4-5.)
Radiated RF IEC 61000-4-3 (alternative method: IEC 61000-4-21)	3 V/m 80 MHz to 2.5 GHz	3 V/m 80 MHz to 2.5 GHz	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$ 80 MHz to 800 MHz (See Table 4-5.) $d = \left[\frac{7}{3} \right] \sqrt{P}$ 800 MHz to 2.5GHz (See Table 4-5.) where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey^a, should be less than the compliance level in each frequency range^b. Interference may occur in the vicinity of equipment marked with the following symbol: 
Note: See on next page for footnotes.			

Table 4-4 Immunity Declaration, cont.

EMC Immunity Guidance & Declaration for EQUIPMENT and/or SYSTEM	
The EQUIPMENT and/or SYSTEM is intended for use in the electromagnetic environment specified in this table. The customer or the user of the EQUIPMENT and/or SYSTEM should assure that it is used in such an environment.	
a	Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the EQUIPMENT and/or SYSTEM is used exceeds the applicable RF compliance level above, the EQUIPMENT and/or SYSTEM should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the EQUIPMENT and/or SYSTEM .
b	Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.
Note: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.	

Table 4-5 Separation Distances

Recommended separation distances between portable and mobile RF communications equipment and the EQUIPMENT and/or SYSTEM .			
The EQUIPMENT and/or SYSTEM is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the EQUIPMENT and/or SYSTEM can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the EQUIPMENT and/or SYSTEM as recommended below, according to the maximum output power of the communications equipment.			
Rated Maximum Output Power (P) of Transmitter Watts (W)	Separation distance according to frequency of transmitter		
	150 kHz to 80 MHz	80 MHz to 800MHz	800 MHz to 2.5 GHz
	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{7}{3} \right] \sqrt{P}$
	Separation Distance meters	Separation Distance meters	Separation Distance meters
0.01	0.12	0.12	0.23
0.1	0.37	0.37	0.74
1	1.17	1.17	2.33
10	3.69	3.69	7.38
100	11.7	11.7	23.3
For transmitters rated at a maximum output power not listed above, the separation distance can be estimated using the equation in the corresponding column, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.			
Note 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.			
Note 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.			

5.3.1 Limitations Management

Adhering to the distance separation recommended in [Table 4-5](#), between 150kHz & 2.5GHz, will reduce disturbances recorded at the image level but may not eliminate all disturbances. However, when installed and operated as specified herein, the system will maintain its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

(*) For example, a 1W mobile phone (800MHz to 2.5GHz carrier frequency) shall be put 2.3 meters apart from the **[EQUIPMENT and/or SYSTEM]** (in order to avoid image interference risks).

5.4 Use Limitation

5.4.1 External components

The use of accessories, transducers, and cables other than those specified may result in degraded ELECTROMATHNETIC COMPATIBILITY of the EQUIPMENT and/or SYSTEM.

5.5 Installation Requirements & Environment Control

In order to minimize interference risks, the following requirements shall apply.

5.5.1 Cable Shielding & Grounding

All interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

5.5.2 Radiated Emission Compliance

This product complies with the radiated emission as per CISPR11 Group 1 Class A standard limits.

The [EQUIPMENT and/or SYSTEM] is predominantly intended for use in non-domestic environments and not directly connected to the Public Mains Network. The [EQUIPMENT and/or SYSTEM] is predominantly intended for use (e.g., in hospitals) with a dedicated supply system and with an X-ray shielded room. In case of use in a domestic environment (e.g., doctors' offices), in order to avoid interferences, it is recommended to use a separate AC power distribution panel and line with an X-ray shielded room.

5.5.3 Subsystem & Accessories Power Supply Distribution

All components, accessories subsystems, systems which are electrically connected to the [EQUIPMENT and/or SYSTEM] must have all AC power supplied by the same power distribution panel & line.

5.5.4 Stacked Components & Equipment

The [EQUIPMENT and/or SYSTEM] should not be used adjacent to or stacked with other equipment; if adjacent or stacked use is necessary, the [EQUIPMENT and/or SYSTEM] should be observed in order to verify normal operation in the configuration in which it will be used.

5.5.5 Low-Frequency Magnetic Field

In case of a digital [EQUIPMENT and/or SYSTEM], the Gantry (digital detector) shall be 1 meter away from the generator cabinet, and 1 meter away from the analog (CRT) monitors. These distance specifications will minimize the low-frequency magnetic field interference risk.

5.5.6 Static Magnetic Field Limits

In order to avoid interference on the [EQUIPMENT and/or SYSTEM] system, static field limits from the surrounding environment are specified.

Static field is specified less than 1 Gauss in Examination room and in the Control Area.

Static field is specified less than 3 Gauss in the Technical Room.

5.5.7 Electrostatic Discharge Environment & Recommendations

In order to reduce electrostatic discharge interference, install a charge dissipative floor material to avoid electrostatic charge buildup.

The relative humidity shall be at least 30 percent.

The dissipative material shall be connected to the system ground reference, if applicable.

Section 6.0

Electro-Magnetic Interference (EMI)

6.1 Gantry and Table

Locate gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.



WARNING REQUISITES FOR ROOM

GANTRY & TABLE ARE ALLOWED TO BE INSTALLED ONLY IN X-RAY PROTECTED ROOMS, WHICH PROVIDE AN ATTENUATION OF AT LEAST 12dB FOR RADIO DISTURBANCES FROM 30 MHz TO 1 GHz.

(The Warning statement above is specified by IEC 60601-1-2, First Edition, 1993)

6.2 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity.

6.3 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

6.4 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics such as console or computer equipment (not including the UPS, which is not classified as sensitive electronics) within any range of the Power Distribution Unit in any direction (including above or below) that would compromise the conditions in [Section 6.1 Gantry and Table](#) above.

6.5 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Healthcare Sales & Service for recommendations. Consider the following when attempting to reduce EMI:

- External field strength decreases rapidly with distance from source of magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single-phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.

Maintain good screening of cables and cabinets.

6.6 UPS

The Uninterruptable Power Supply (UPS) provides a consistent power supply to various electrical components of the system. Also, it continues to provide electrical power to components during a site-wide power outage so components can be safely shut down. Do not place sensitive electronics such as console or computer equipment (not including the PDU, which is not classified as sensitive electronics) within any range of the Universal Power Supply in any direction (including above or below) that would compromise the conditions in [Section 6.1 Gantry and Table](#) above.

Chapter 5

Floor Loading and Weights

Section 1.0 Floor Loading

The Non-Fixed LightSpeed 5.X system has a total floor load of approximately 6500 lbs (2950 kg). About 5185 lbs (2350 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly.

Table 5-1 lists CT components with weight, size, floor loading and normal mounting requirements.

ITEM	NET WEIGHT LB(KG)	OVERALL W X D INCH (MM)	WEIGHT/AREA LB/SQ. FT. (KG/M ²)	LOAD PATTERN IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
Gantry (no covers)	3770 (~1710)	86.6 X 39.4 (2200 X 1000)	140 (682)	Rectangular base plate 24 X 81 (610 X 2057) with four round pads, each 2.5 (63.5) in contact with floor. Individual pad loadings are: 910 lb, 960 lb, 1040 lb, and 1090 lb (see Figure 5-3).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at four leveling pads into concrete floor.
Dollies (each)	250 (114)				
Top Cover (each)	24.6 (11.2)				
Side Cover (each)	25 (11.3)				
Front Cover	95 (43)				
Rear Cover	100 (45)				
Patient Table	1185 (538) Includes 450 (204) Patient	24.3 X 94 (617 X 2387)	175 (852)	Rectangular base 17 X 57.5 (432 X 1460) with five round pads, each 2.5 (63.5) in contact with floor. Worst-case cantilever pull on any bolt is 1235 lb (561 kg).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at three leveling pads into concrete floor.
Dolly	215 (98)				
Power Distribution Unit	~770 (~350)	28 X 22 (711 X 559)	185 (909)	Four Casters support area of 28 X 22 (711 X 559).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets ² in seismic zones.
Console w/HP & w/o monitors	450 (204)	48 X 39 (1219x991)	140 (681)	Four Casters or Leveling Feet support area of 46 X 19 (1168 X 483).	Casters are for positioning. Set on floor. Console may be anchored to floor using angle brackets ²
Monitor - LCD (ea)	22 (10)				

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel. [See statements in 8.2 - Table and Gantry Mounting Requirements, on page 30.]
- 2.) Non-fixed unit manufacturer will provide appropriate shock mounting for PDU.
- 3.) Non-fixed unit manufacturer will provide appropriate mounting procedure, using existing mounting hardware, for Operator's Console.
- 4.) Non-fixed unit manufacturer will install supplied shock mounting kit for UPS.

Table 5-1 LightSpeed 5.0 System Floor Loading

Section 2.0

Mobile Mounting Data

Note: It is the van/unit manufacturer's responsibility to provide an approved support structure and mounting method. General Electric is not responsible for any failure of the support structure or method of anchoring.

The following pages show center-of-gravity information for system components:

- Table: [Figure 5-1](#)
- Gantry: [Figure 5-2](#)
- Power Distribution Unit: [Figure 5-4](#)
- Operator's Console/Computer: [Figure 5-5](#)

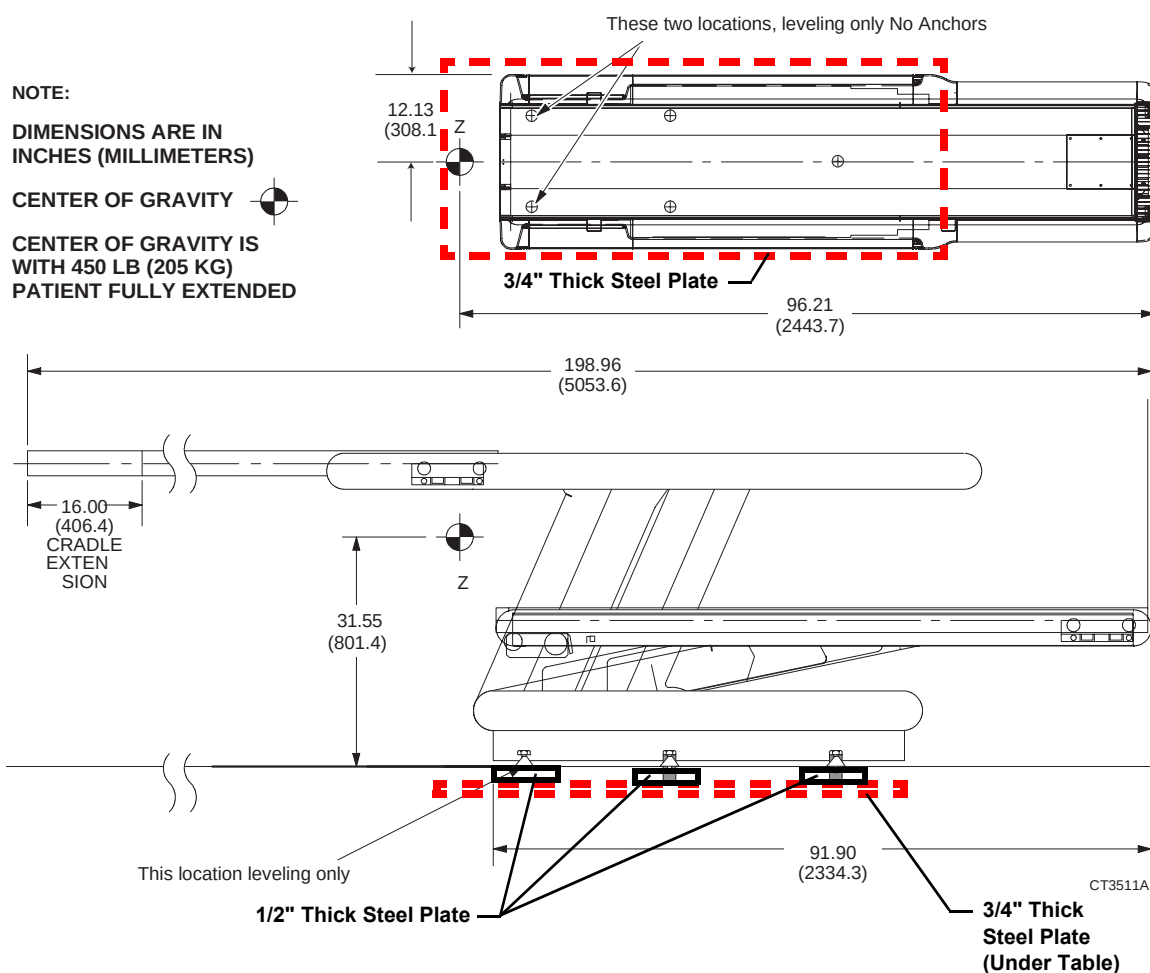


Figure 5-1 Patient Table (CT1)

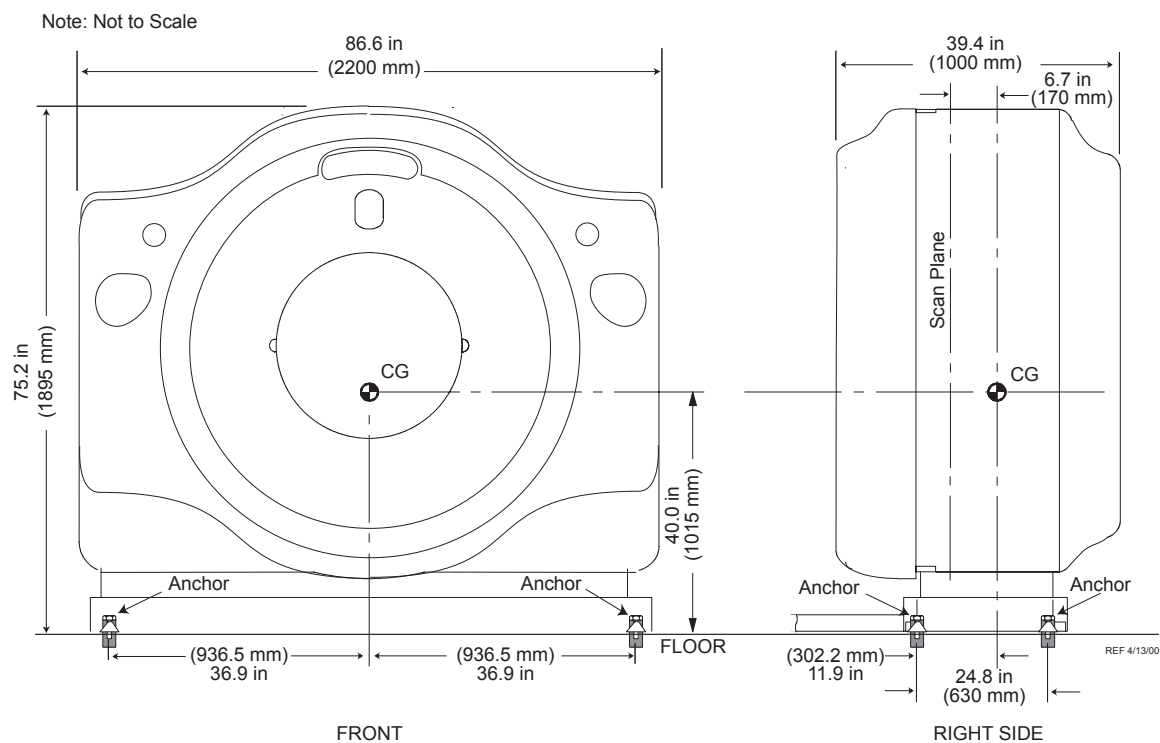


Figure 5-2 Gantry (CT2)

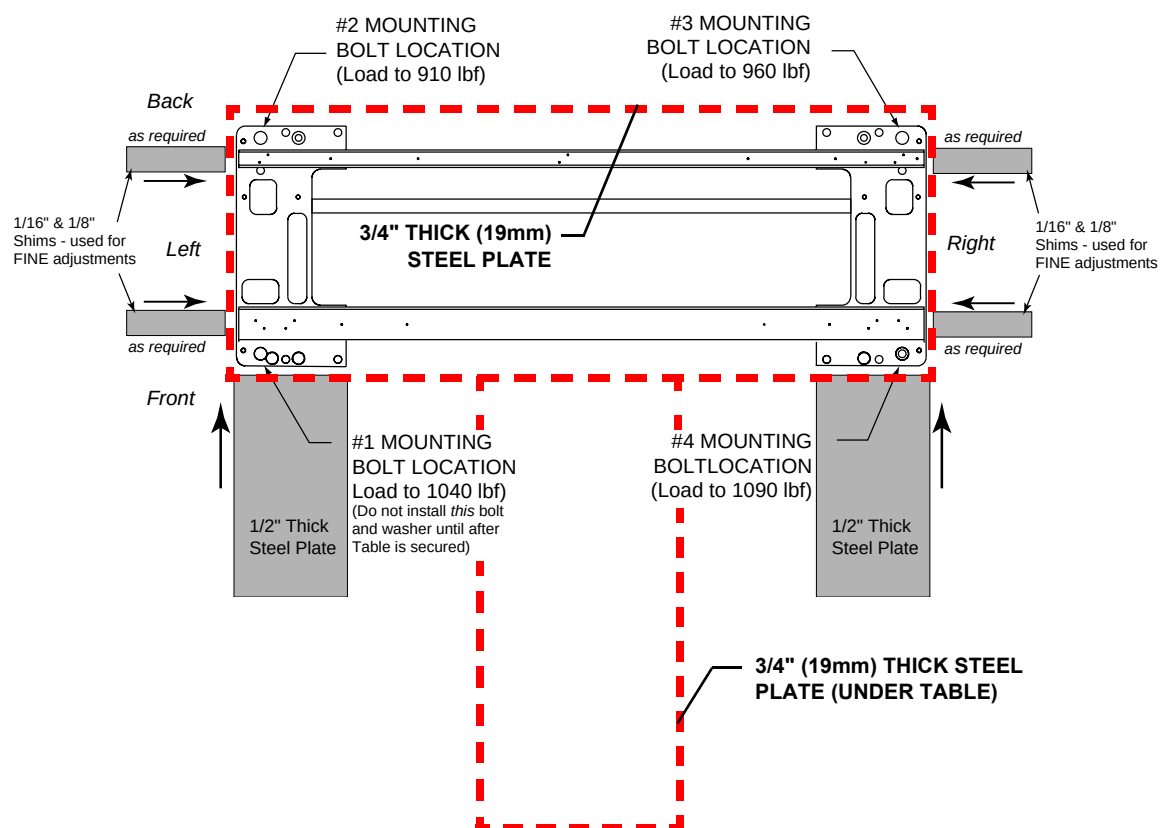


Figure 5-3 HP Gantry Base Mounting and Steel Shim Plates - Top View

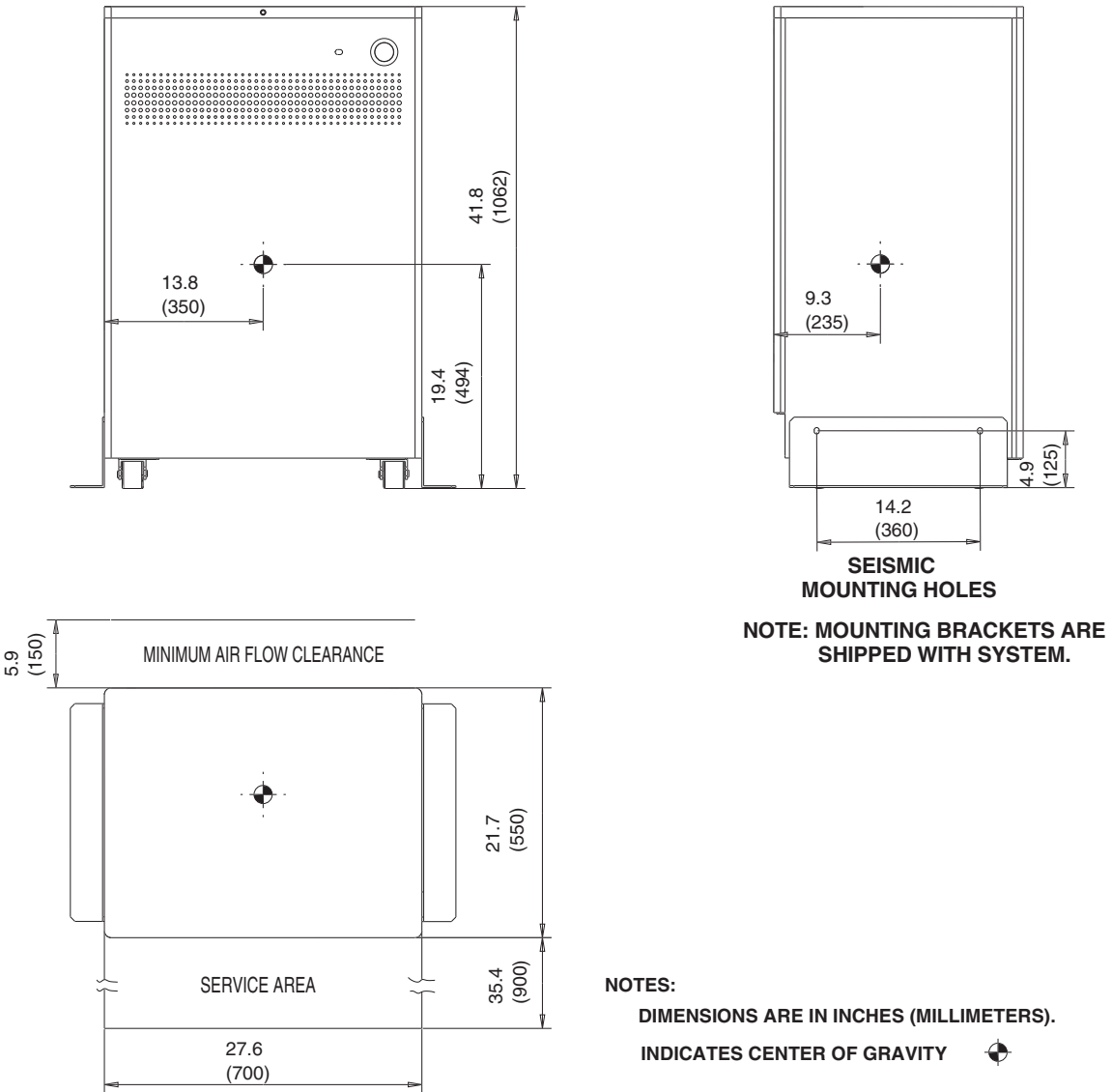


Figure 5-4 Power Distribution Unit (NGPDU)

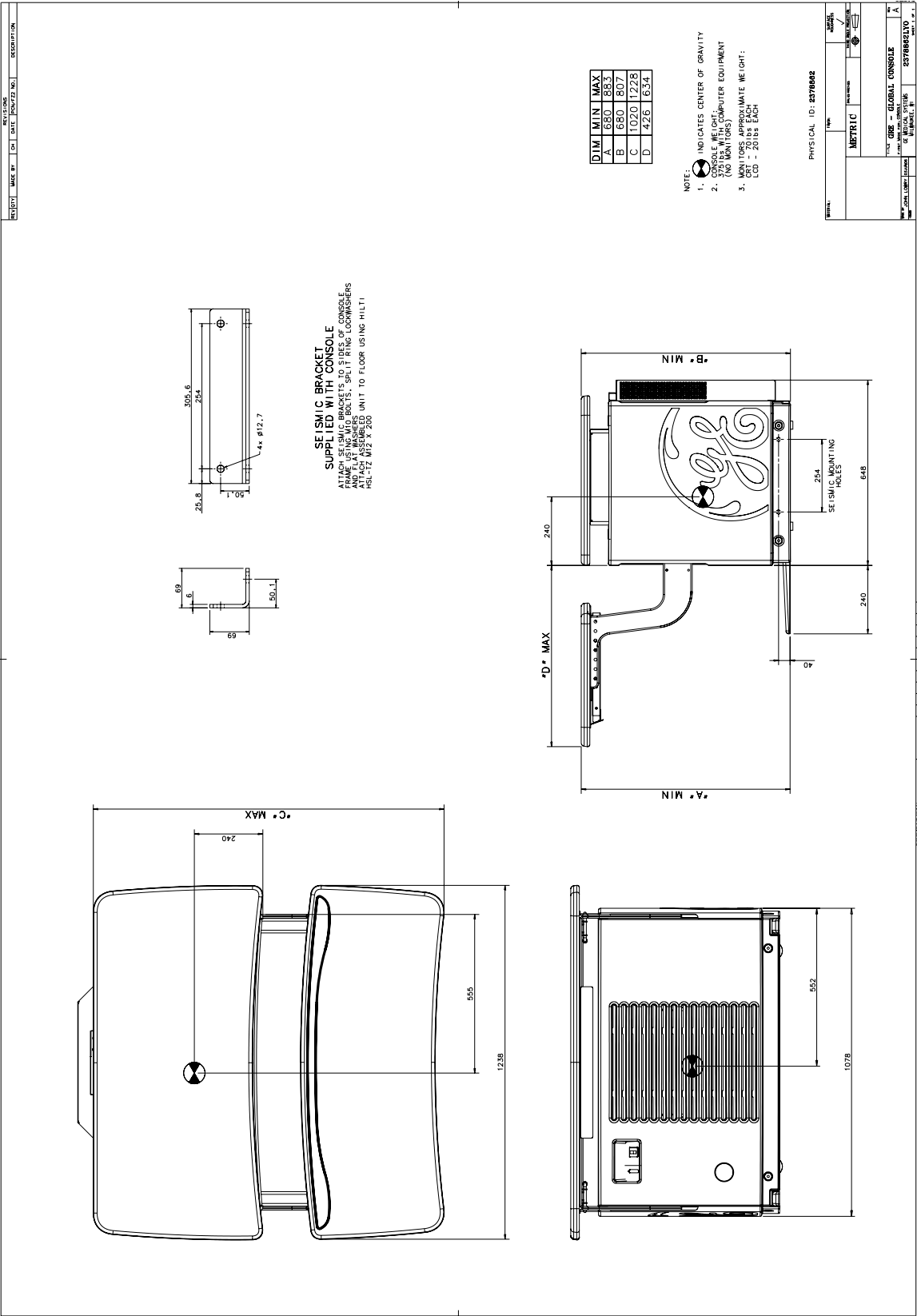


Figure 5-5 Operator's Console

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Chapter 6

Delivery Data

SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0

Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 20 shipping containers, which include dollies, skids and boxes without skids.

Section 2.0

Delivery/Shipping Requirements & Considerations

The LightSpeed system is not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping or hoisting.

The Gantry, Console, Table and PDU must NEVER be dropped. A drop from a height greater than one half inch ($\frac{1}{2}$ ") may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the Gantry, Dock to Dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, the system may be moved via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces.

When moving Gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the Gantry at the slowest reasonable rate. See [Figure 6-1](#).

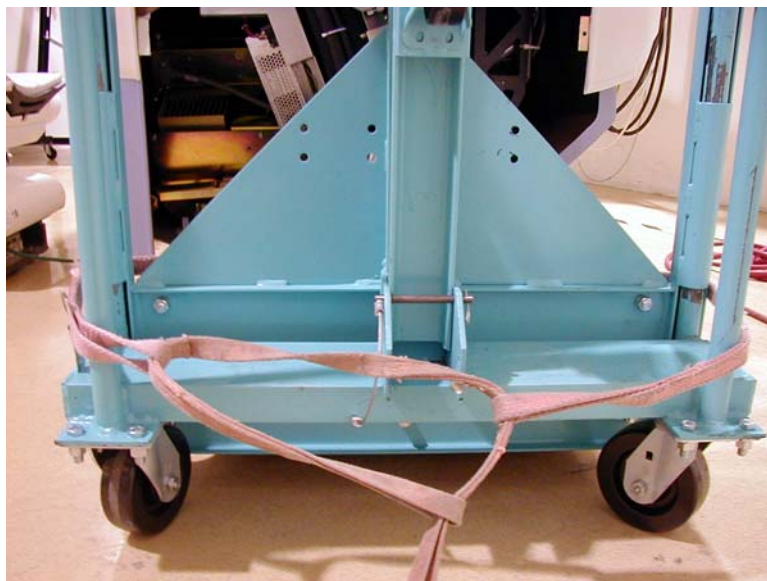


Figure 6-1 Gantry Strap Location

The LightSpeed system—including Gantry, Console, Table and PDU—is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the Console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive or computer failure) may not be evident until after system installation is complete.

All system components must remain upright at all times, and must not be tipped. Nor should the Gantry be hoisted. The Gantry should be moved by rolling it on its dollies only. Movement through hallways, doorways, elevators, etc., must be done without tipping or lifting the Gantry.

Protection for flooring along the move path (from dock to scan room) is advised.

Section 3.0

Site Environmental Considerations

3.1 Dust/Dirt Contamination

LightSpeed systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.



NOTICE The act of installing a GE CT scanner in a construction (i.e., unfinished) site will likely result in the following adverse effects:

- Increased installation time
- Decreased installation quality
- Increased scanner downtime, due to increased service calls

3.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed CT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer’s requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration’s should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0

Storage Requirements

SHORT TERM STORAGE (LESS THAN 6 MONTHS)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 40° to +80° F (4° to +27° C).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum relative humidity rate of change is 5%/hr.
- The maximum temperature rate of change is 5° F/hr. (3° C/hr.)
- Air pressure should be between 700hPa and 1060hPa.

**NOTICE**

Between delivery is considered short-term storage.

Van storage must meet the same specifications as above.



Figure 6-2 Package Symbols (Storage)

LONG TERM STORAGE (6 MONTHS OR MORE)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 50° to +90° F (10° to +32° C).
- Maintain relative humidity (non-condensing) between 20% and 70%.

Section 5.0

Extreme Temperature Transportation and Deliveries

Extreme temperatures should be avoided during system transportation and delivery.

Extreme temperatures are defined as below Zero degrees Fahrenheit (-18° C) or above 120 degrees Fahrenheit (49° C), without humidity control.

Section 6.0

System Transportation

When transporting the CT system, ensure that the system is not exposed, for an extended period of time, to temperatures or humidity outside of the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 0% to 80%

**NOTICE**

Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

Section 7.0

Gantry Considerations

The gantry is shipped with most covers installed. The assembly is mounted between two dollies. See [Figure 6-3](#). Two side rails are bolted to the dollies to stabilize dollies and protect gantry. Use dolly elevating casters to lift gantry off its base and roll it into position.



Figure 6-3 Gantry with Shipping Dollies and Side Rails

Door Openings. Clear door openings for moving equipment into building must be 42 X 82 in. (1067 X 2083 mm) minimum. 8 ft. (2439 mm) corridor width is helpful.

Elevator requirements. To move gantry from receiving location to scanning room, consider elevator capacity and size. By removing side rails and one dolly after gantry is placed in elevator, gantry width/length and elevator depth requirements are reduced. Contact a representative of elevator manufacturer if gantry weight exceeds elevator capacity.

CONFIGURATION	LENGTH IN (CM)	WIDTH IN (CM)	HEIGHT IN (CM)
Dollies On, Side Rails On	114 (290)	51 (129)	79 (200)
Dollies On, Side Rails Removed	114 (290)	39.4 (100)	79 (200)
Dollies Off, Covers Off	81 (206)	34 (86)	73 (185)

Table 6-1 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door size for gantry with covers and dollies attached, but side rails removed, is 40 inches (101.6 cm).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Dollies: Typically, dollies are used on the gantry, table, and console for domestic shipments. Once the gantry, table and console are installed at the site, return dollies to GE using the shipping document located in Box #1.

Dollies can be purchased for international shipments (B7850LD) to be used at the customer site. After the system has been removed from the crates, dollies shipped with international shipments **only** are not to be shipped back to GE in Milwaukee, WI, USA, but to be retained to the local GE office or warehouse.

Zero clearance dollies are available (P/N 2128404).

Section 8.0

Table Considerations

The table is shipped without side covers installed. Covers are shipped in four separate boxes. The table is mounted between two dollies.

Table shipping dimensions are 101" (2565 mm) long, 32" (81 mm) wide, and 47" (119 mm) high.

Section 9.0

Console Considerations

NEW INFORMATION -- PLEASE REVIEW THIS SECTION

The console is shipped without the keyboard table installed. The keyboard table is shipped with the console.

- The console is shipped on a skid equipped with ramps for unloading.
- Do not remove the console from the shipping skid until it is in the CT equipment room.
- Console shipping dimensions (on the shipping skid) are 40" (1016 mm) deep, 53" (1346 mm) wide, and 41" (1041 mm) high.

Note: The dimensions of the console alone (as shipped) are 34.95" (888mm) deep, 48.74" (1238mm) wide, and 29.5" (750mm) high.



NOTICE Do not lift the console by the monitor table top to remove from the shipping skid.



Figure 6-4 Console ready to be unloaded from shipping skid.

Chapter 7

Power Requirements

Section 1.0

Introduction

Note: Power requirements for LightSpeed 5.X differ from those for LightSpeed 4.X and previous systems.

The power distribution unit (PDU) supplied with the LightSpeed system transforms and distributes power to all system components. The PDU is the only power entry point required to operate system. To minimize voltage regulation effects, power wiring between the facility main distribution panel and the PDU should be kept as short as possible.

When routing the power wiring all three phase wires and ground must be run in the same conduit or raceway duct. Power wires should be routed separately from system control and signal cables, using a separate conduit or trough in raceway duct.

Metallic conduit, floor duct or surface raceway may be used for running cables, depending upon local codes and practices. However, cable passageways should be large enough to install any cable with all other cables already installed. Use of non-metallic conduit is not recommended.

Section 2.0

System Input Power - Van Manufacturer

2.1 Van Source

Power to the system should be supplied by a dedicated feeder from the nearest Main Distribution Panel (MDP). Refer to [Table 7-1](#) to determine wire size. A 1/0 ground is always required.

A protective disconnect device is provided.

2.2 Facility Source

- Power to the system should be supplied by a dedicated feeder from the nearest Main Distribution Panel (MDP) within the facility serving the non-fixed unit.
- A protective disconnect device must be provided in the facility serving the non-fixed unit.

2.3 Main Disconnect Control

Van manufacturer supplied emergency off buttons are to be mounted in the CT Room near each exit, 60 in. (1424 mm) from the floor and connected to the protective disconnect device, in order to disable the power to all CT system equipment in emergency situations. This button is to be clearly labeled "Emergency Off" and visible to personnel in the CT Room. It is important that the button be labeled "OFF" and not "STOP". An additional emergency button should be mounted in the computer/equipment room near the exit door.

GE supplied main disconnect controller (MDC) is lockable to meet OSHA requirements for power Lockout/Tagout.

2.4 Configuration

The Mobile LightSpeed system is designed to operate on three-phase, **four-wire** wye power. A ground referenced wye source produces the lowest leakage currents and is preferred. However, the neutral wire does not need to be run to the system, i.e., four-wire connection. (A dummy terminal is provided for 'parking' the neutral wire in the event a five-wire service is already installed at the site.)

2.5 Rating

The LightSpeed 5.0 system operates on three-phase power meeting the following specifications.

Voltage	380 to 480 VAC
Capacity	90 kVA
Frequency	50 or 60 Hz +/- 3 Hz

- Maximum power demand = 90 kVA @ 0.85 PF at a selected technique of 140 kV, 380 mA.
- Average (continuous) power demand at maximum duty cycle = 20 kVA.

The "A1" disconnect device referenced above must provide over-current protection for the system and facilitate multi-point remote "Emergency Off" control of system power. A disconnect utilizing under-voltage release control is preferred over shunt trip devices. The rating of the "A1" disconnect device depends on the nominal line voltage at the site. Refer to [Section 3.0: "Recommended Power Distribution System"](#), for minimum rating requirements and suggested disconnect devices.



NOTICE The electrical rating is described on the system rating label attached on the gantry; not on the PDU.

2.6 Regulation

Total load regulation as measured at the PDU input terminals must not exceed 6%. The capacity of the facility transformer and size & length of feeder wires directly affect the load regulation presented to the system. Refer to [Section 3.0: "Recommended Power Distribution System"](#), for recommended single-unit installation specifics.

2.7 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.8 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in [Table 7-1](#). Maximum transient voltages should be limited to 1500V peak.

2.9 Grounding

Metal conduit, raceway or the armor of armored cable used to power the system should be bonded to the PDU cabinet. However, in addition to such mechanical grounding, a dedicated 1/0 (55 mm²) or larger insulated copper ground wire must be run with the phase wires from the main distribution panel to the PDU.

Note: The shield or armor of armored cable is not sufficient for this purpose.

The ground wire should be bonded to intermediate distribution panels through which it passes in accordance with local codes. The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and earth must not exceed 2 ohms.

Section 3.0

Recommended Power Distribution System

All power sources must meet the minimum power requirements. The power distribution system requirements are applicable to the internal wiring of the van, the on-board generator and the external wiring at each facility that uses the van.

This means that on-board generator power shall meet all of the facility specifications of a fixed site.

The minimum recommended transformer size is 112.5 KVA, rated 3.2% regulation at unity power factor. The minimum recommended feeder size and over-current protection device ampacity based on line voltage are shown in [Table 7-1](#).

- Maximum power demand = 90 kVA @ 0.85 PF at a selected technique of 140 kV, 380 mA.
- Average (continuous) power demand at maximum duty cycle = 20 kVa.
- Maximum allowable total source regulation = 6%.

	EUROPE	USA
Nominal Line Voltage (VAC)	400	480
Voltage Range (VAC) +/- 8%	368-432	442-518
Average Line Current (A)	29	24
Momentary Line Current (A) @ Nominal Line Voltage	130	108
Maximum Line Current (A) @ Low Line Voltage	141	117
Primary Disconnect Device (A1) Over-current Protection (A) ⁶	110	90
Feeder Length (MDP to A1) ¹		
0-100 ft (0-30.5 m)	2 (35) ³	3 (30) ³
100-250 ft (30.5-76 m)	2 (35) ³	3 (30) ³
251-300 ft (77-91 m)	1/0 (55) ³	2 (35) ³
301-350 ft (92-106 m)	2/0 (70) ³	1 (45) ³
351-400 ft (107-122 m)	3/0 (85) ³	1/0 (55) ³
Sub-Feeder Length (A1 to PDU)		
0-40 ft (0-12 m)	2 (35) ²	3 (30) ²

Notes:

- 1.) The feeder table above is based on the use of copper wire, rated 75C and run in steel conduit. Wire size is in AWG (mm²). Ampacity is determined in accordance with the National Electric Code (NFPA 70), Table 310-16 (1999).
- 2.) The wire size shown is the minimum allowable for the specified over-current protection rating.
- 3.) The feeder is sized to contribute 2.4% maximum regulation.
- 4.) A minimum 1/0 (55 mm²) ground wire is required.
- 5.) A neutral wire is not required and, if present, is not used.
- 6.) The Primary Disconnect (A1) must have provision for tagout/lockout. [Figure 8-6, on page 74](#), shows the primary power disconnect (A1 - P/N R4502AD) using a combination fusible disconnect and magnetic contactor.

Table 7-1 Facility Power Requirements

The non-fixed unit manufacturer will communicate system power requirements.

If the Mobile LightSpeed system must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment that use rapid film changers should be avoided.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to Mobile LightSpeed meet all the requirements.

Section 4.0 Uninterruptable Power Supplies

Uninterruptable Power Supplies (UPS) are standard on Mobile LightSpeed systems.

Section 5.0 Power Audit (Facility Only)

A facility site power audit is required for the Mobile LightSpeed family of products. This site power audit can be arranged with the GE Power Quality team.

Section 6.0 Ground System

Mobile LightSpeed has been designed to use an equal potential grounding system. The required ground system is shown in [Figure 7-1](#). There are three primary grounding points:

- A system power ground point located in the PM (Power Module).
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

All exposed metal surfaces in patient vicinity are grounded to the reference ground point.

For additional information, refer to Electrical Safety Equipment, Direction **46-014505**.

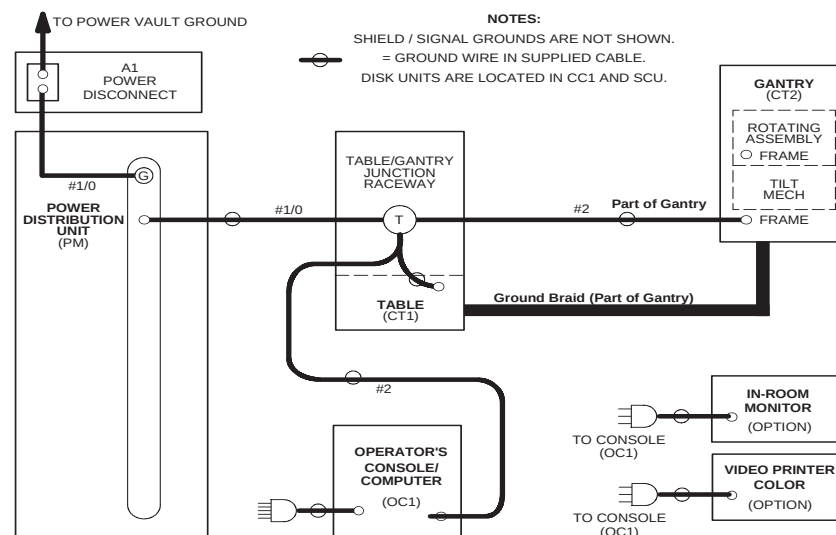


Figure 7-1 System Ground Map

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Chapter 8

Interconnection Data

Section 1.0

UPS Interconnect

UPS Catalog Numbers:

- B7999PP - HP Gantry (LS 5.X)
- B7999PS - H16 Gantry (LS4.x) and H3 Gantry (LS3.X)

Refer to Mobile Installation documentation included with the UPS system, to install this product.

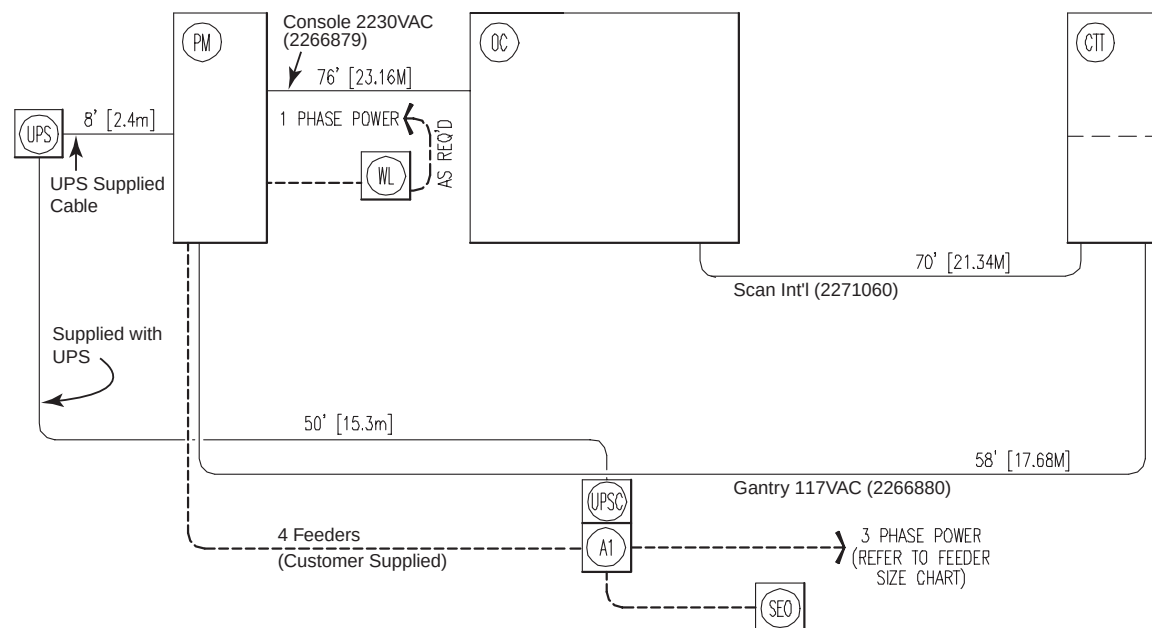


Figure 8-1 Typical UPS Interconnect



Figure 8-2 Typical UPS

Section 2.0 Interconnect Runs, Wiring and Cables

2.1 GEMS Supplied - LightSpeed 5.X LS16, Ultra, Plus (Optional, Long Run)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	63 (55)	19.3 (16.76)	2343529	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (.87) Dia
051	62.5 (55)	19 (16.76)	2343530	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15.3 (.604)	4	14	11.2 (.44) Dia
052	60 (55)	18.5 (16.76)	2343528	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
053	80 (75)	24.5 (22.86)	2343531	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12.3 (.483)	3	10	56.4 (2.22) Dia
055	63 (55)	19.3 (16.76)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
056	83 (75)	25.5 (22.86)	2371450-3	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	63 (55)	19.3 (16.76)	2333152	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
101	83 (75)	25.5 (22.86)	2333150	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24.3 (22.86)	2373436-1	Signal (LAN), Gantry to OC			1900	<30VDC		5.9 (.234)	8	24	15 (.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (.39) Dia

Table 8-1 GEMS Supplied Cables for LightSpeed 5.X LS16, Ultra, Plus (Optional, Long Run) - UL Information

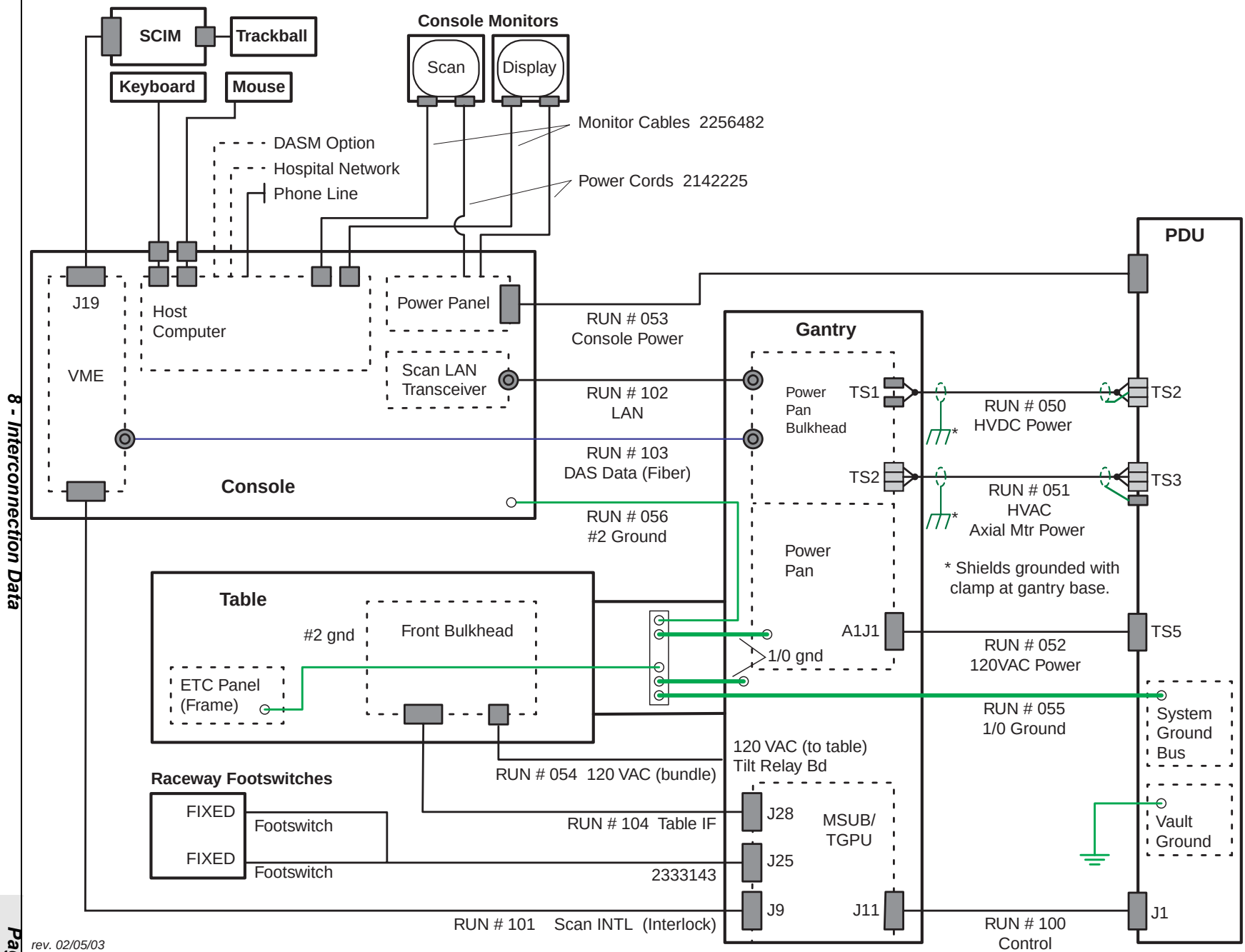


Figure 8-3 System Interconnect Diagram (LightSpeed 5.X LS16, Ultra, Plus)

2.2 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M.)	
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M.)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2-1,2,3,4							
RUN NO. 5 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							
*	REFER TO Table 8-3 on page 72 FOR AWG (MM2) WIRE SIZES.								

Table 8-2 Runs 1, 2, 3, 4 and 5 Connections

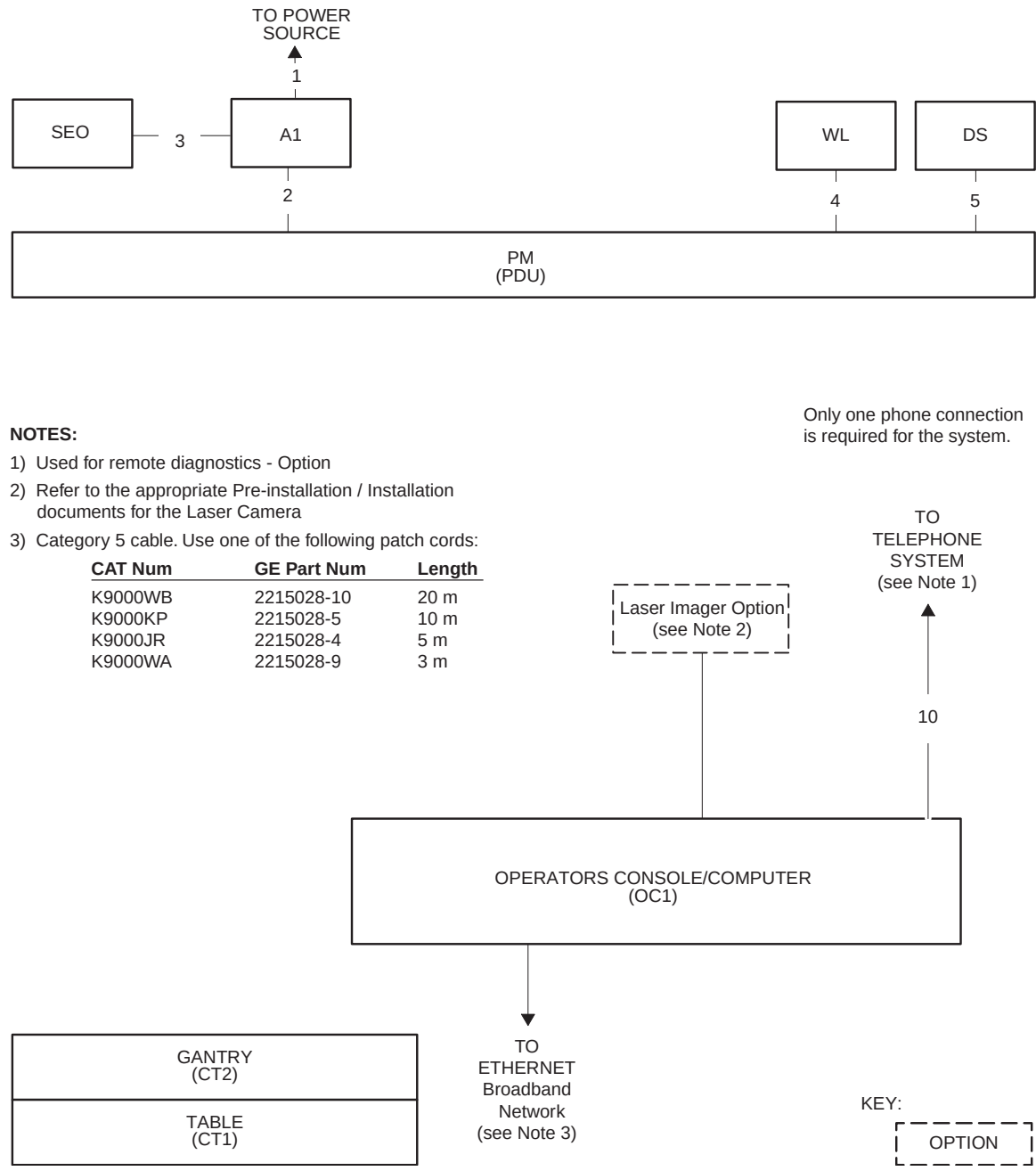


Figure 8-4 Interconnection Runs

2.3 Van Manufacturer Supplied Cables

MANUFACTURER SUPPLIED WIRING		DESCRIPTION	WIRE & CABLE PIGTAILS FT. (M)	
QTY	SIZE AWG (MM ²)		FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1) Maximum Run Length *				
3	*	Power	3 (1)	3(1)
1	1/0 (50)	Ground	3 (1)	3 (1)
1	*	Neutral	3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *				
3	*	Power	3 (1)	3(1)
1	1/0 (50)	Ground	3 (1)	3 (1)
1	*	Neutral	3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)				
2	14 (2)	Power	6 (2)	6 (2)
1	14 (2)	Ground	6 (2)	6 (2)
RUN NO. 8 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)				
2	14 (2)	Warning Light 24V Control A3J2-1,2,3,4		
RUN NO. 9 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)				
2	14 (2)	Scan Room Door Interlock A3J6-1,2		
RUN NO. 11 OPERATOR'S CONSOLE TO LASER IMAGER (OPTION)				
1		Power		
1		Network		
RUN NO. 12 OPERATOR'S CONSOLE TO ETHERNET BROADBAND NETWORK				
1		Network cable with two RJ45 connectors		
RUN NO. 13 GANTRY TO GANTRY TILT WARNING LIGHT (VAN EXTERIOR)				
1		Pair of wires from gantry base, supplied by mfg.		

Table 8-3 Manufacturer Supplied Cables

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY VAN MANUFACTURER	USA VENDOR / CAT NO. GE CATALOG
TTL	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by facility that uses the van.	

Table 8-4 Additional Manufacturer-Supplied Components

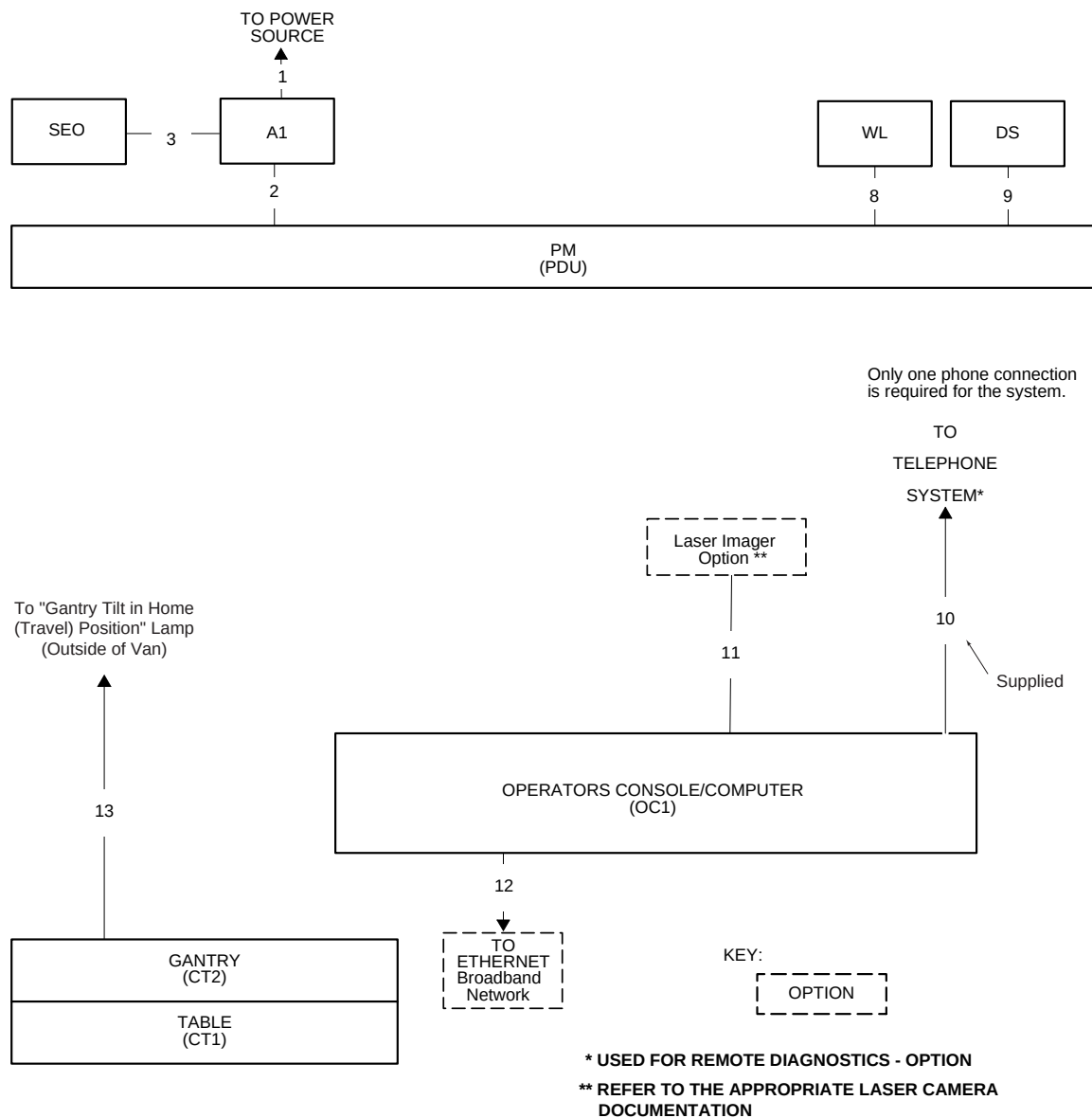


Figure 8-5 Interconnection Runs

Section 3.0

Typical Van Manufacturer-Supplied Wiring

3.1 Primary Power Disconnect

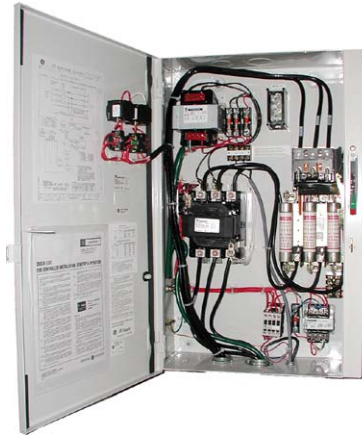


Figure 8-6 Primary Power Disconnect (A1)

3.2 Scan Room Warning Light & Door Interlock (NGPDU)

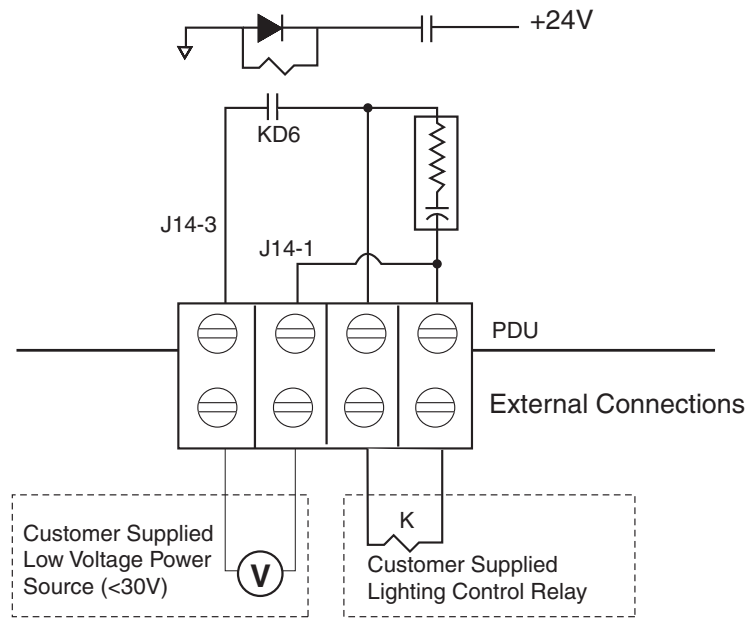


Figure 8-7 Typical Scan Room – Warning Light Connection to PM

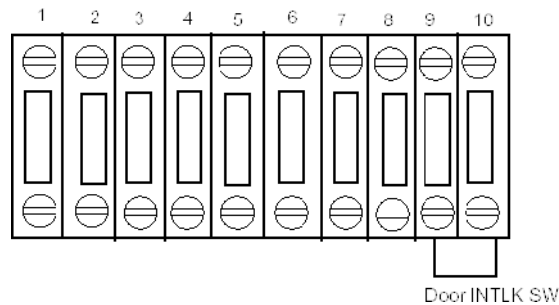


Figure 8-8 Typical Door Interlock

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